

Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 271 factors, ordered by in-sample L–S CAGR.

HOW TO READ

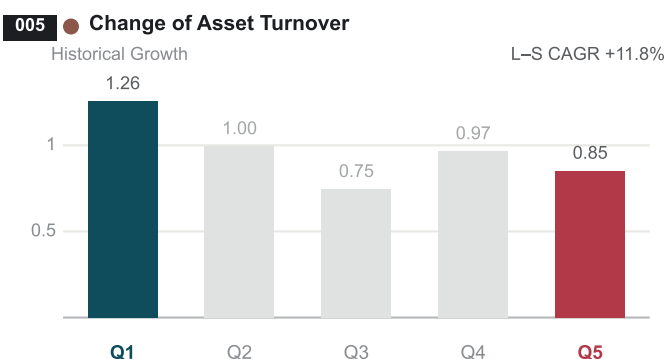
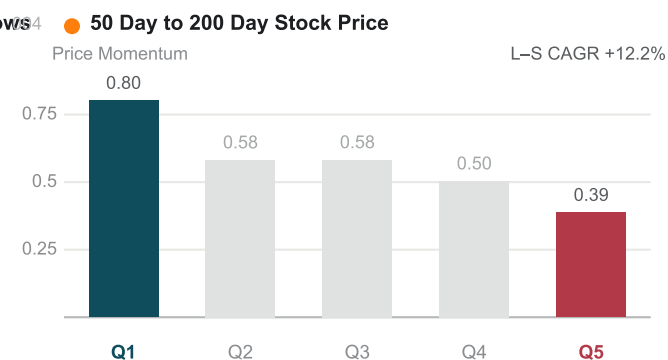
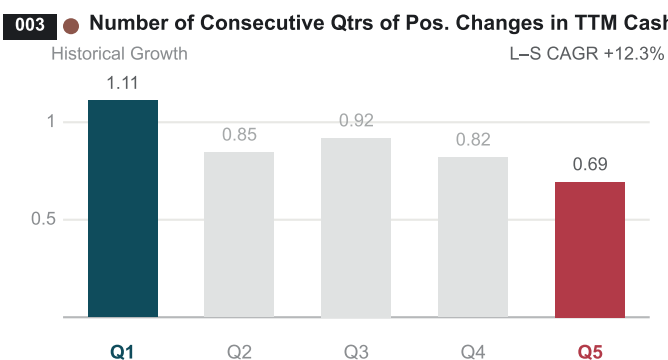
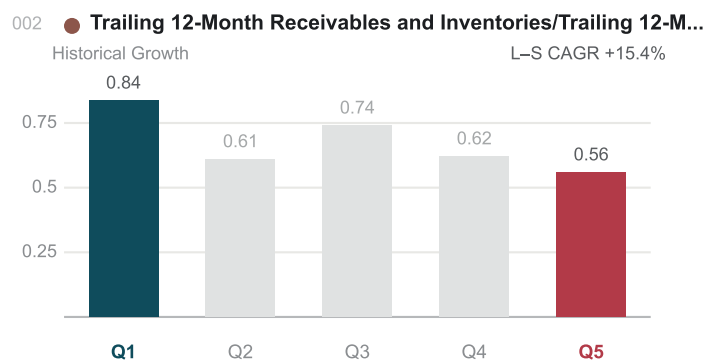
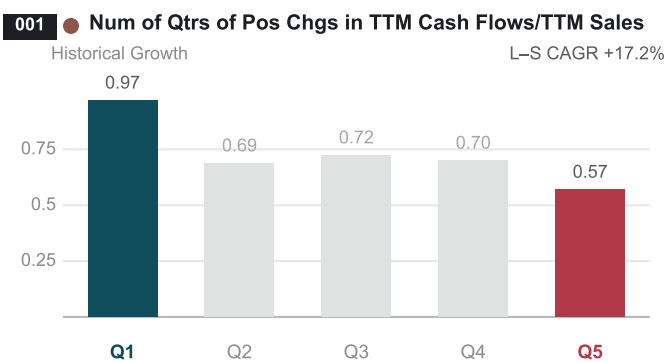
 Long  Short  Neutral

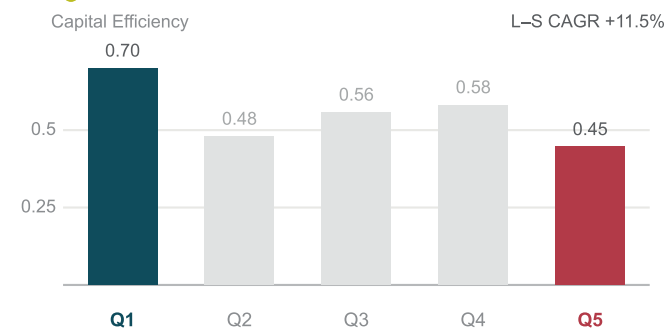
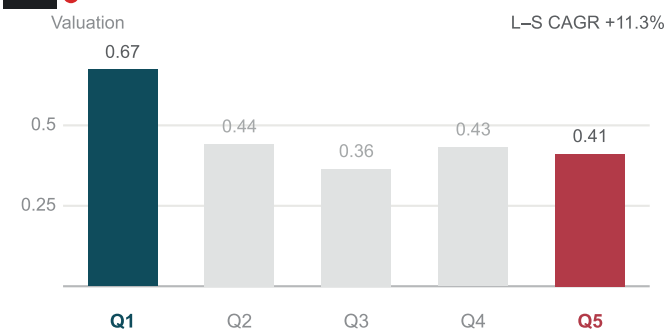
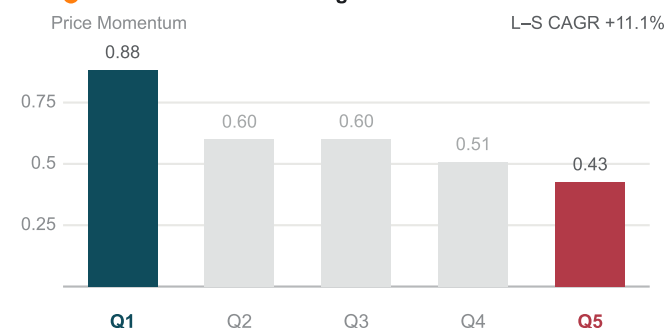
Quintiles are selected as long or short by performance against a spread-based threshold — not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

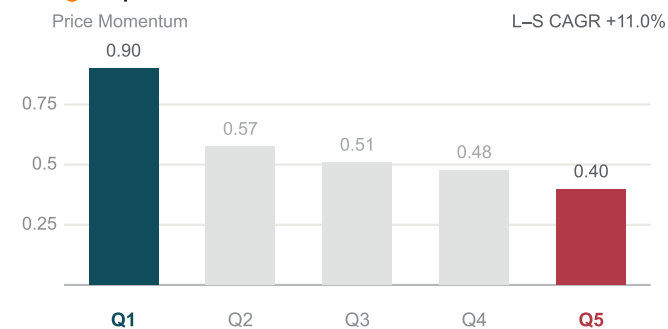
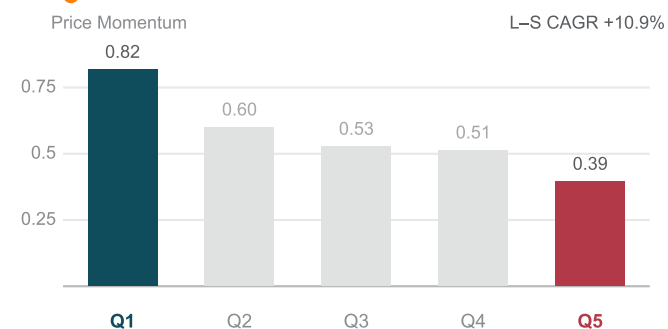
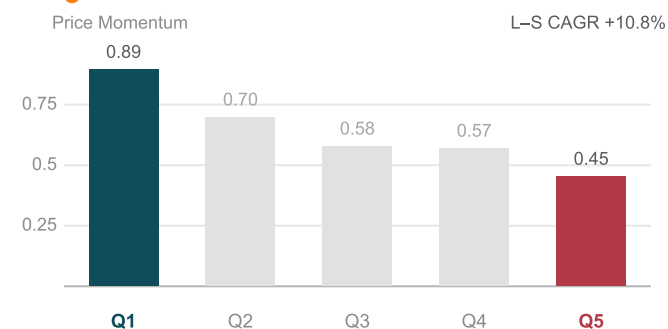
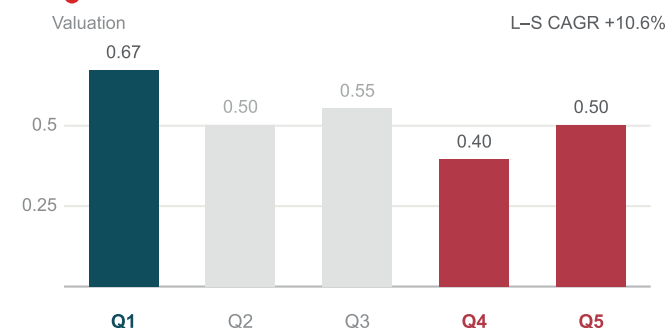
FACTOR STYLES

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 Earnings Quality	55	 Size	1
 Analyst Expectations	24	 Historical Growth	56
 Capital Efficiency	28	 Volatility	10

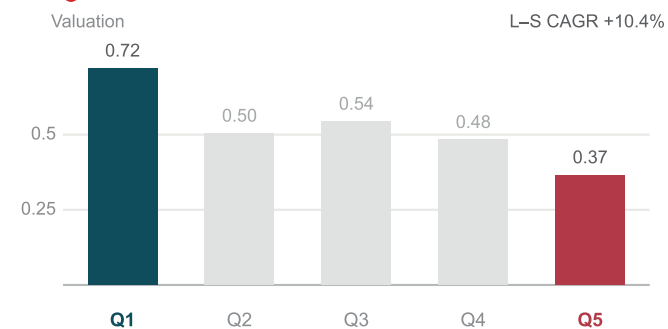
 001 In current model portfolio (50 factors)



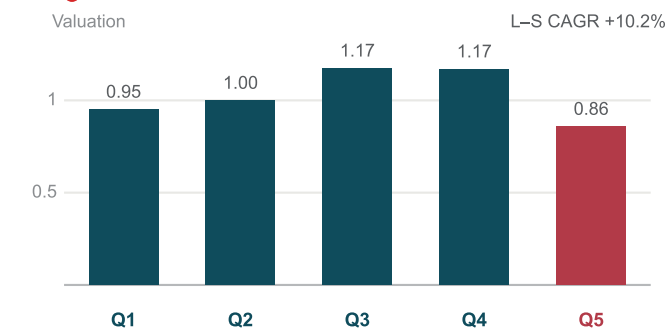
007 ● **Tobin's Q Ratio**008 ● **39W Lagged Return**009 ● **Pretax Income Per Share to Price**010 ● **Operating Earnings to Price Ratio**011 ● **26W Relative Price Strength**012 ● **9 Month Price Momentum**

013 ● **12M Realized Price Volatility**014 ● **Slope of 52W Price Trend**015 ● **15/36 Week Price Ratio**016 ● **4W to 52W Price Oscillator**017 ● **Analyst Dispersion for FY2 EPS**018 ● **Book to Price**

019 ● Normalized EP



020 ● Indicated Annual Dividends to Price Ratio



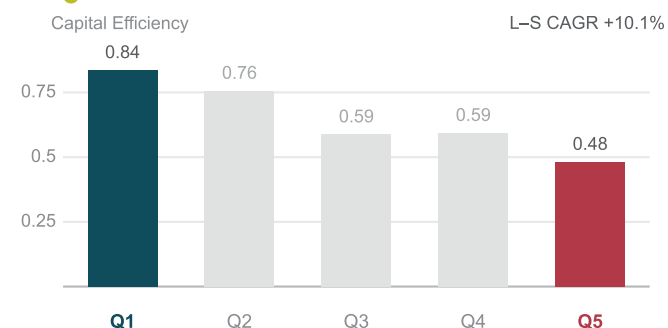
021 ● Operating Earnings to Assets Ratio



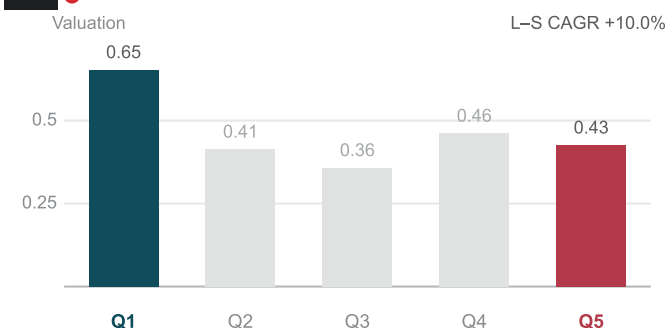
022 ● Percent Above 260-Day Low



023 ● Return on Assets



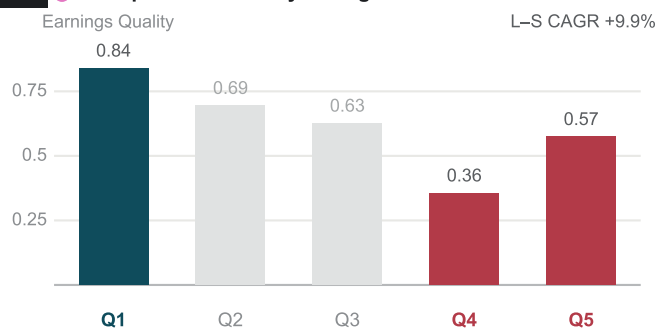
024 ● EBITDA to Price



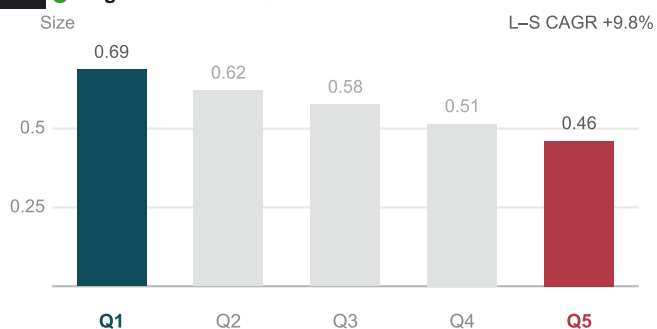
025 ● Stability of Change in Number of Holders



026 ● Unexpected Inventory Change



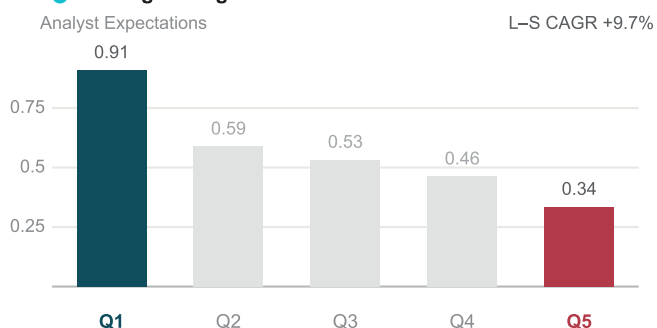
027 ● Log of Total Last Quarter Assets



028 ● Assets to Price Ratio

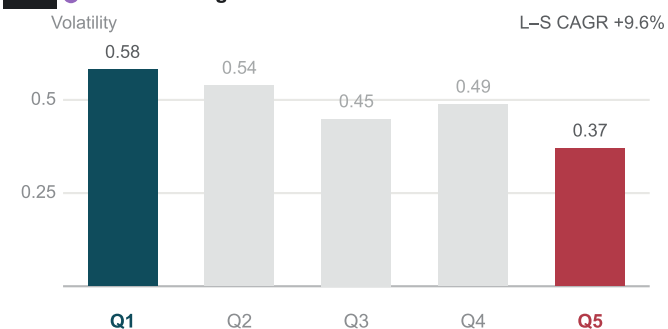
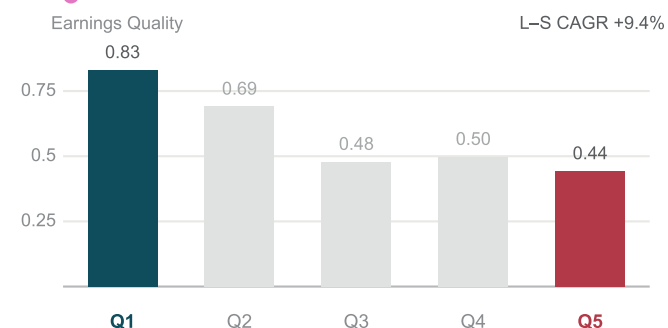


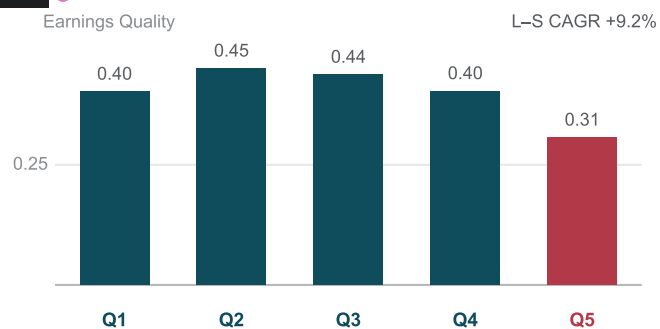
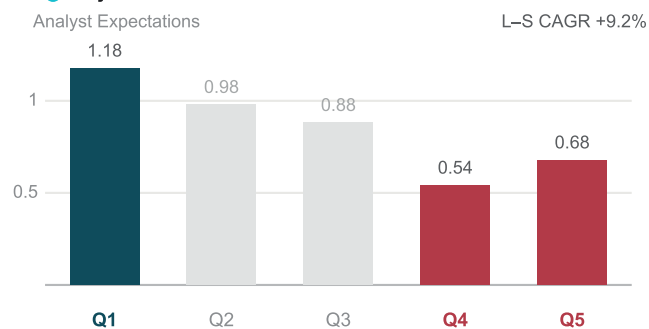
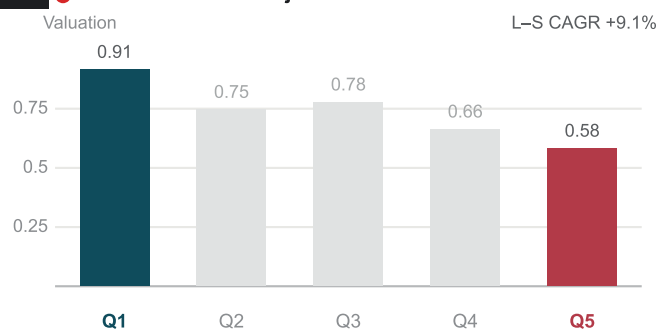
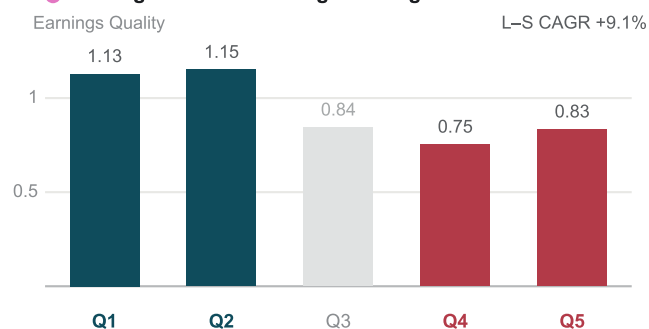
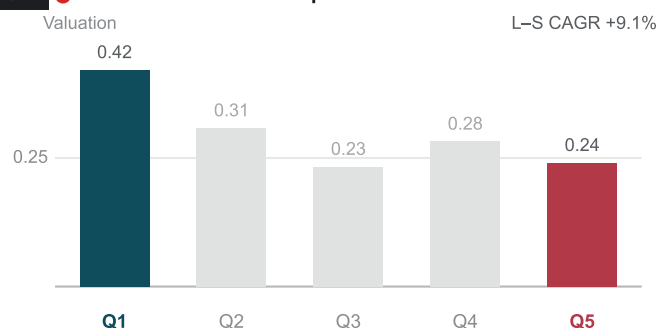
029 ● 6M Chg in Target Price



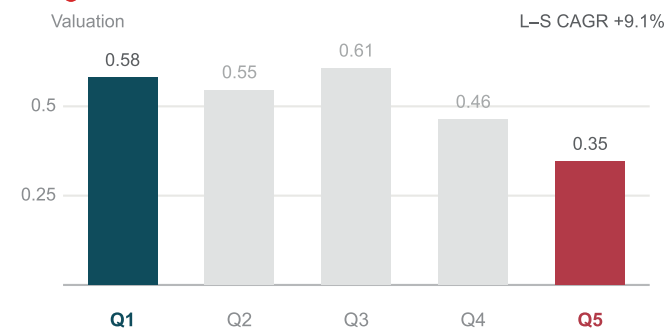
030 ● Cash to Price



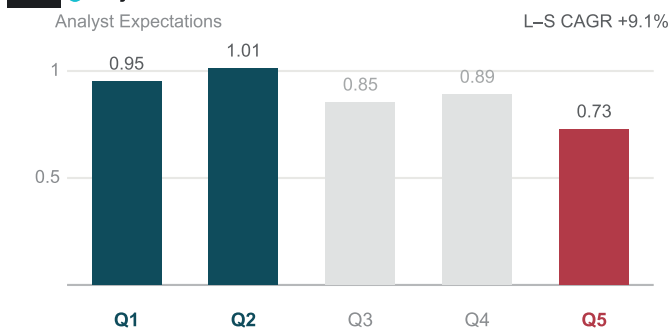
031 ● Sales to Price Ratio**032 ● Growth Flow to Price****033 ● Book Leverage****034 ● Earnings to Price****035 ● Cash Flow to Total Asset****036 ● Number of Consecutive Qtrs of Positive Changes in TTM Quart...**

037 ROA 20 Qtr Standard Deviation**038 Adjusted 3M Revision in FY1 EPS Estimate****039 Inverse P/E Ratio Adj for Growth and Yield****040 Trailing 12-Month Earnings/Trailing 12-Month Sales****041 Net Cash Flow to Enterprise Value****042 Solvency Ratio**

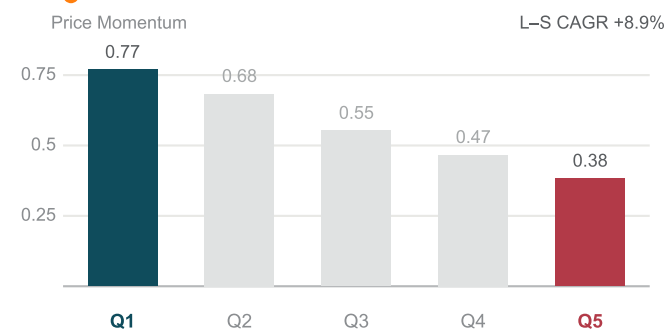
043 ● Dividends to Price Ratio



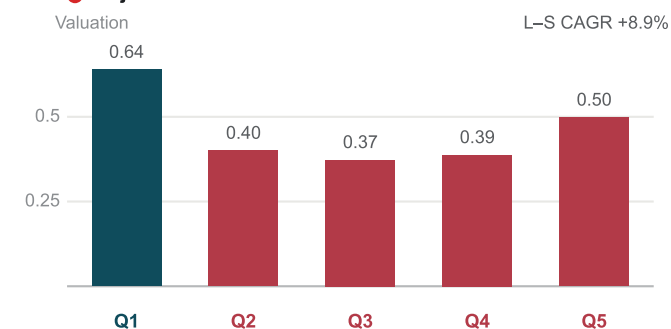
044 ● Adjusted Inverse PEG



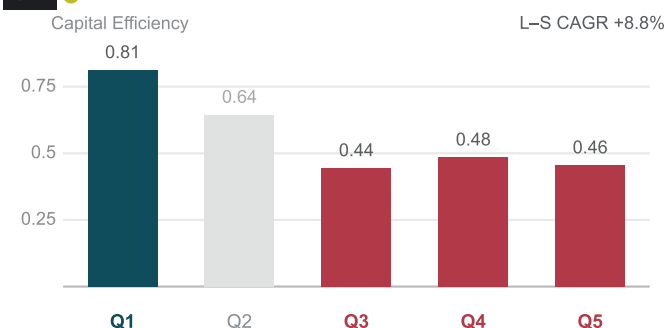
045 ● 180D Price TStat



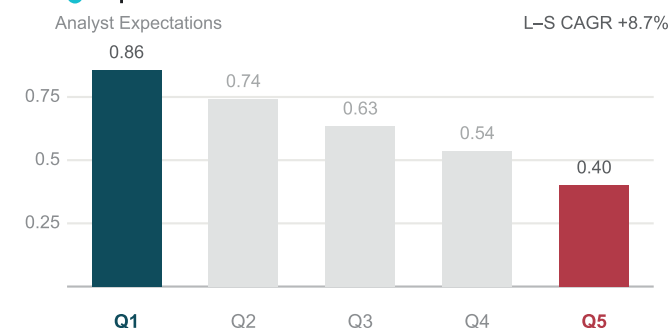
046 ● Adjusted EBIT to Price

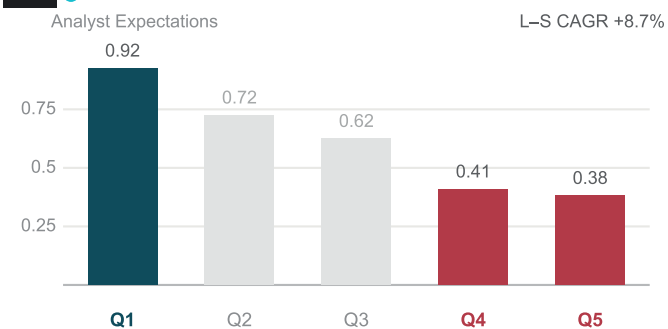
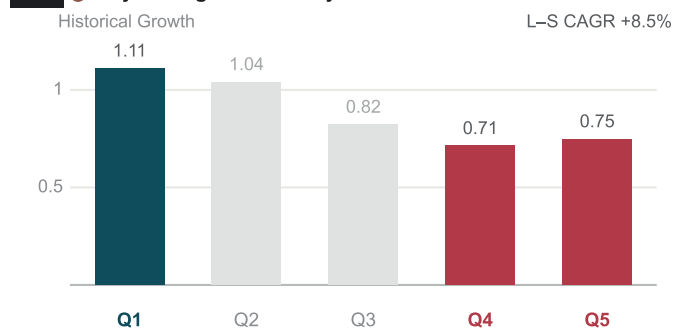
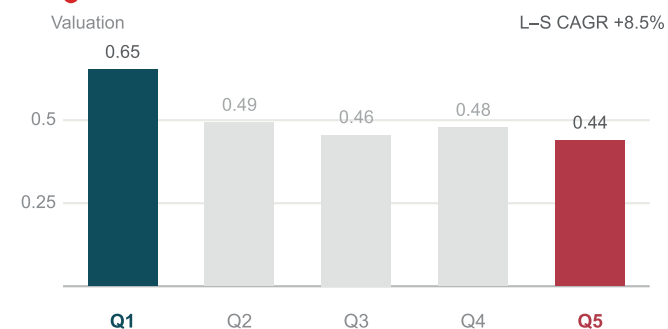
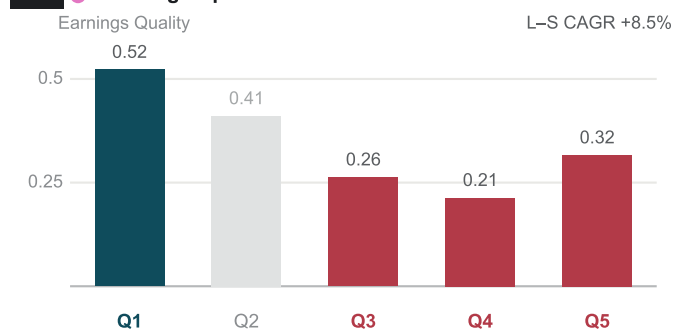
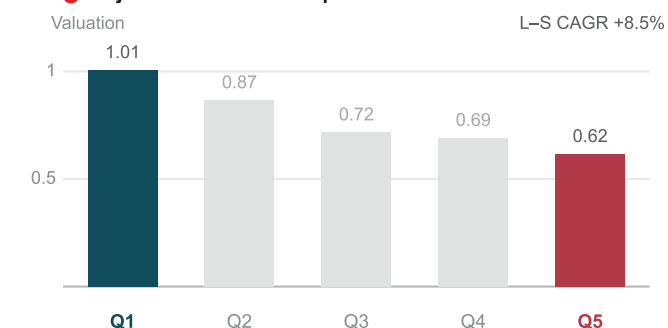
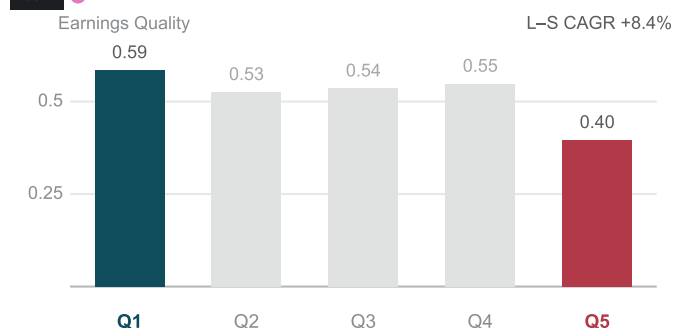


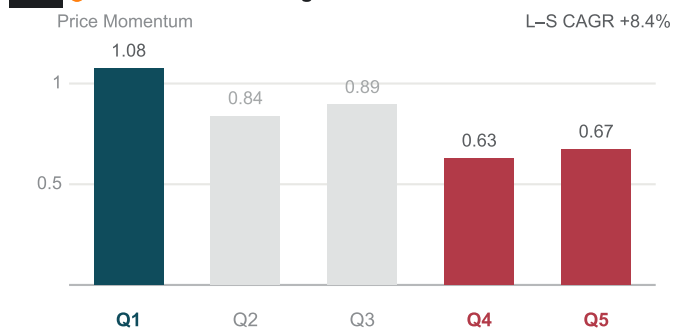
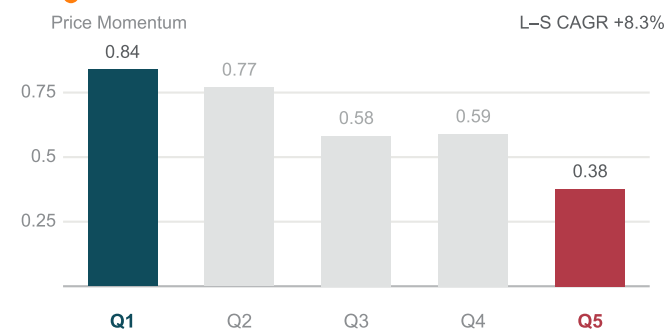
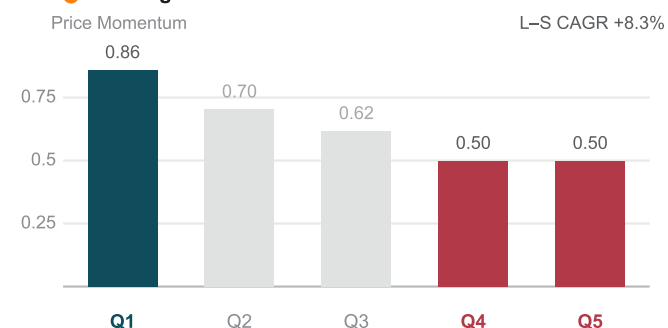
047 ● EBITDA to Assets



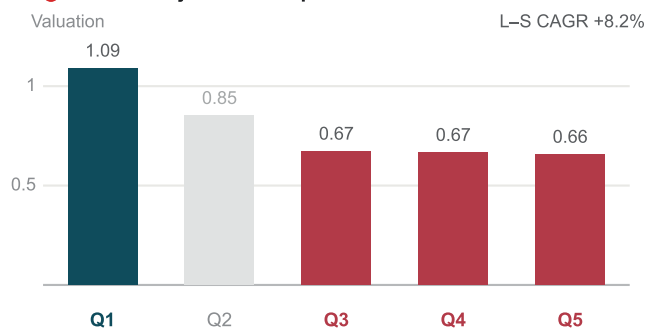
048 ● Expected LTG



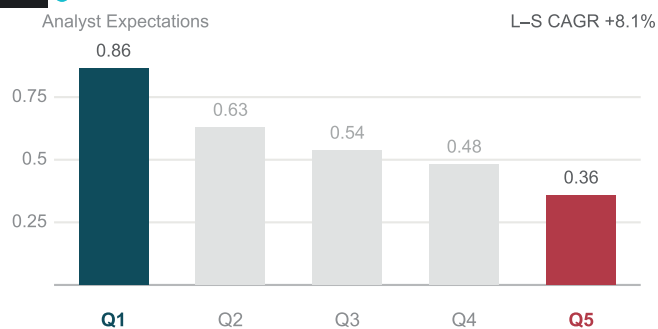
049 3M Revision in FY1 EPS Estimate**050 Adj 1Y Chg in Asset Adjusted FCF****051 Sales to EV Ratio****052 Working Capital to Sales****053 Adjusted Book to Enterprise Value****054 Balance Sheet Accruals**

055 ● EBITDA to Enterprise Value**056 ● Short Interest Trading Volume****057 ● 12M - 1M Price Momentum****058 ● Amihud Illiquidity Measure****059 ● 52W High Low****060 ● 52-Week Volume Price Trend with 20-Day Lag**

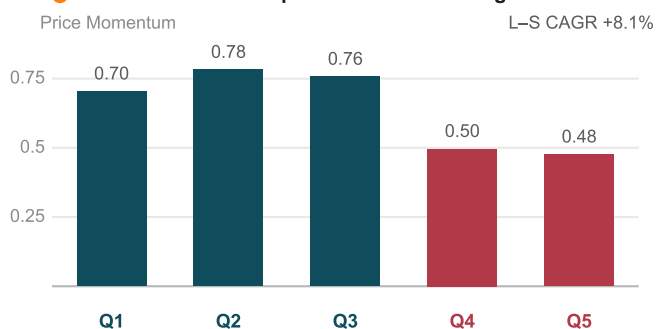
061 ● Cash to Adjusted Enterprise Value



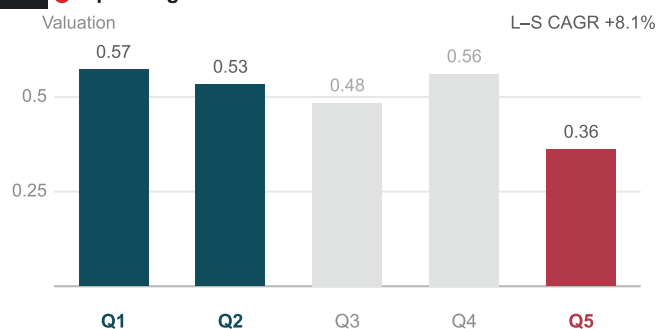
062 ● 3M Revision in FY2 EPS Estimate



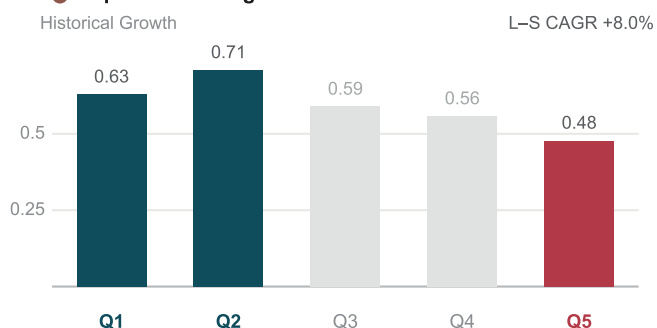
063 ● Institution Ownership Level - Active Manager



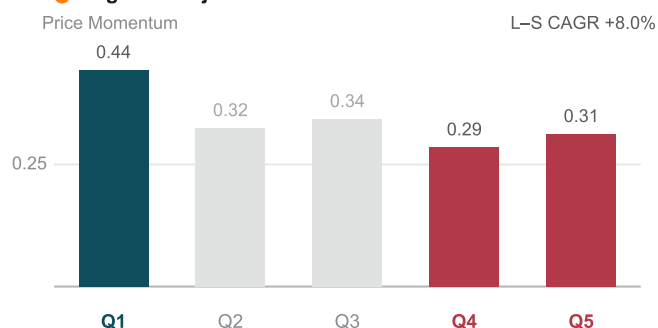
064 ● Operating Cash Flow to Price

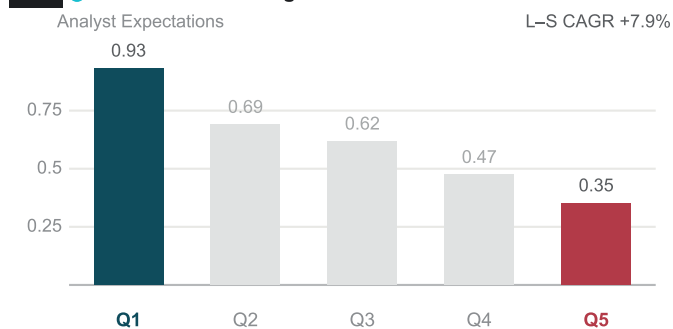
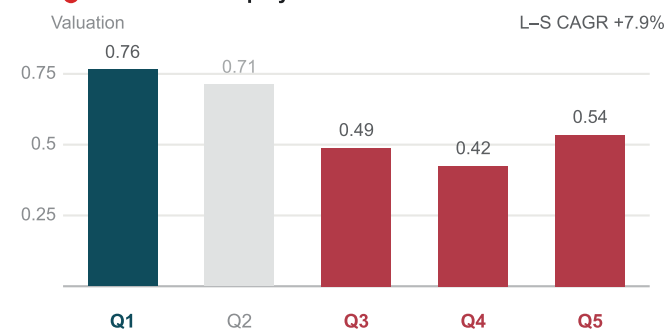
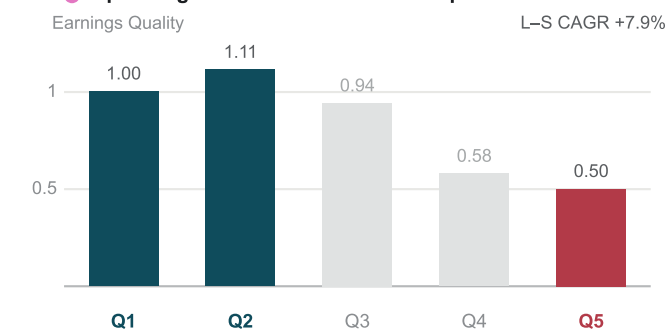
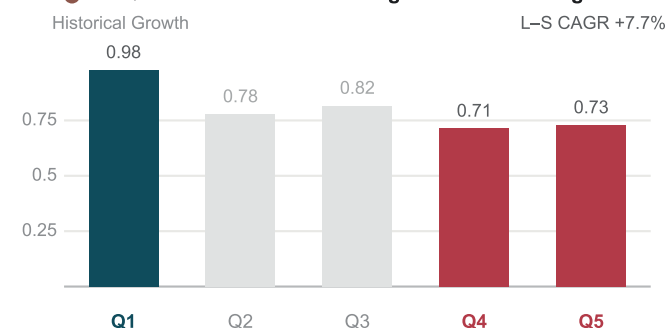


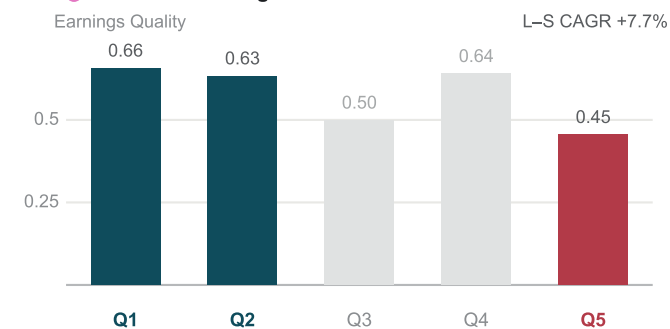
065 ● Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1



066 ● Log of Unadjusted Stock Price



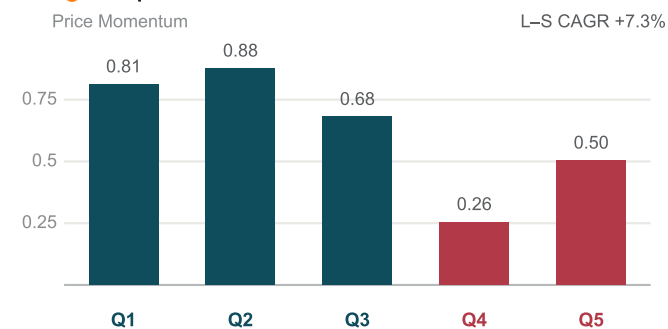
067 ● 3Y Chg in Operating Cash Flow to Price**068 ● Street Revision Magnitude****069 ● Cash Flow to Equity****070 ● Operating Cash Flow to Invested Capital****071 ● Forward Earnings to Price****072 ● 12-Quarter Trendline in Trailing 12-Month Earnings**

073 ● **Adjusted Forward Earnings to Price**074 ● **Gross Profit Margin**075 ● **60M CAPM Beta**076 ● **Adjusted 3Y Chg in Asset Adjusted Free Cash Flow**077 ● **Adjusted Revision Magnitude**078 ● **1Y Chg in Asset Adjusted EPS**

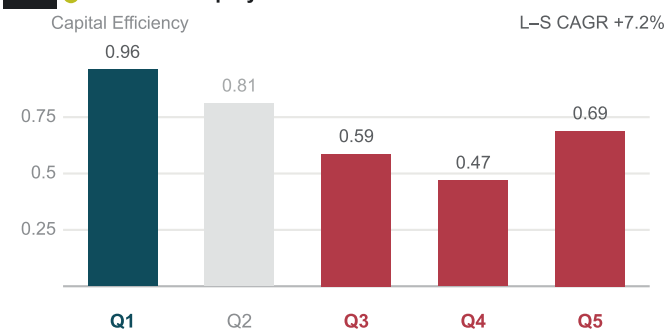
079 ● Net Cash Flow to Price



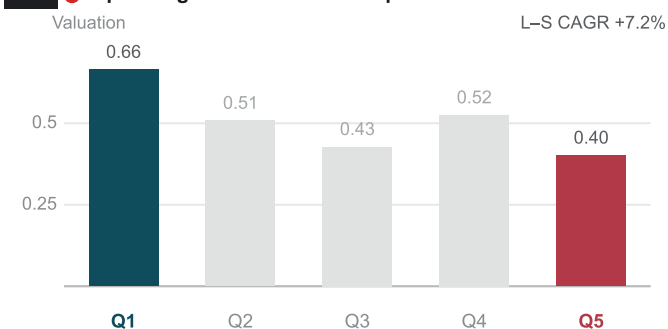
080 ● Sharpe Ratio



081 ● Return on Equity



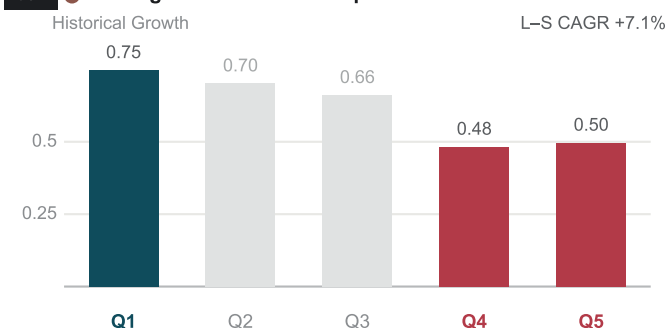
082 ● Operating Cash Flow to Enterprise Value

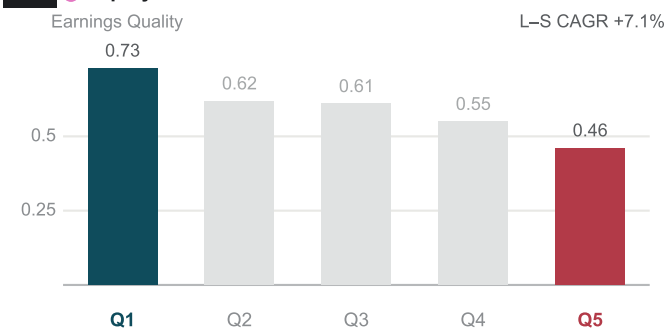
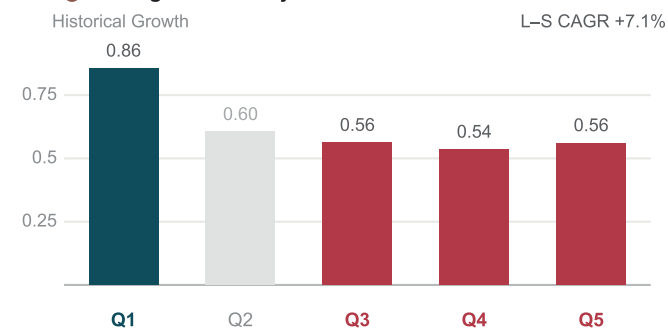
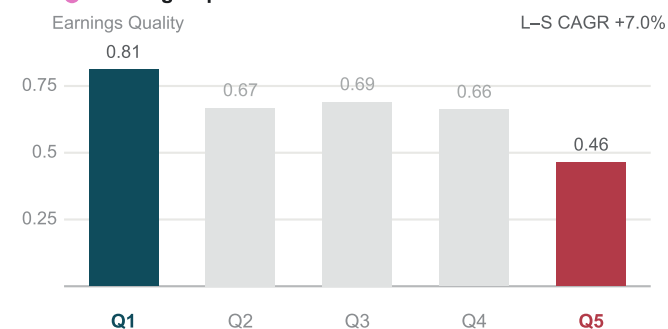
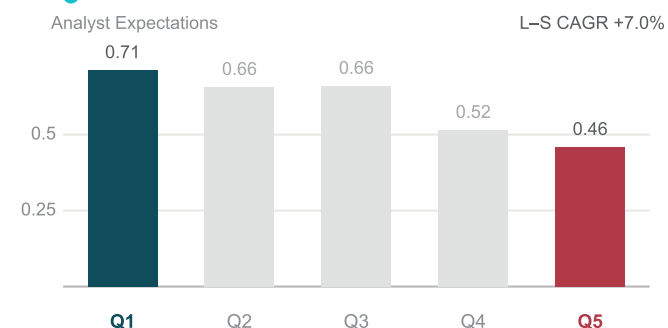


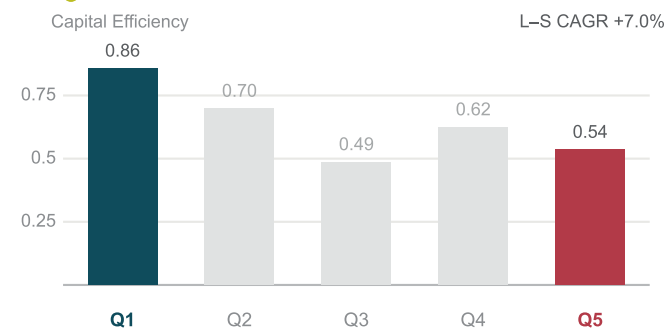
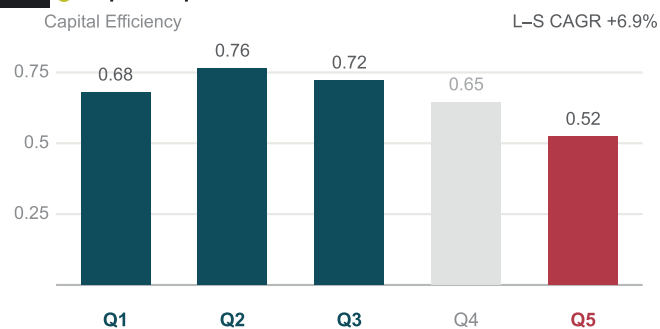
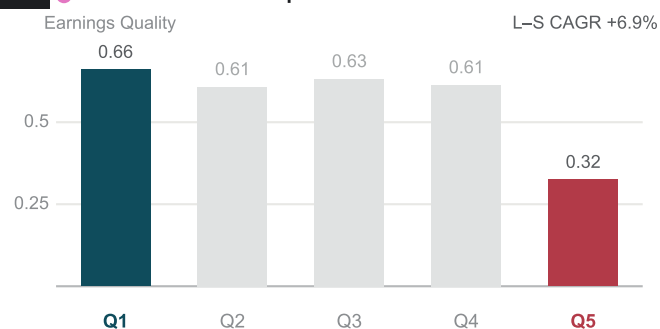
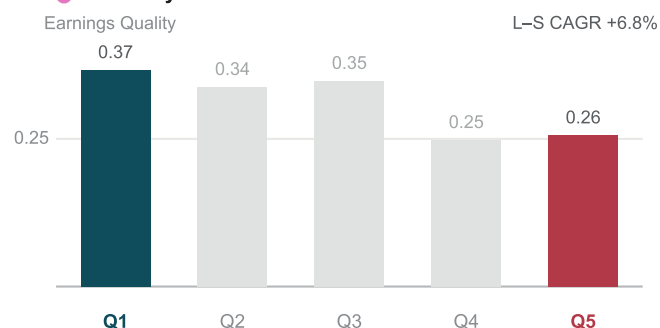
083 ● 1Y Chg in Operating Margin



084 ● 1Y Chg in Free Cash Flow per Share



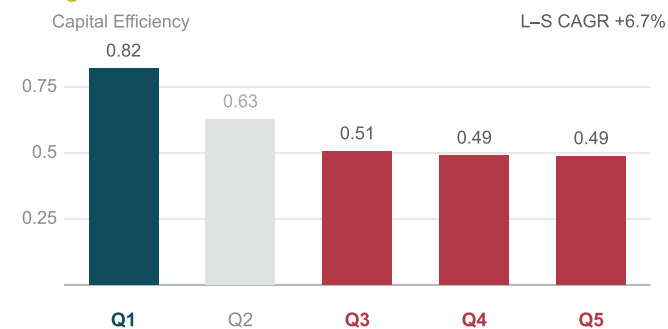
085 Equity Turnover**086 1Y Chg in Asset Adjusted Cash Flow****087 90 Day Coefficient of Variation****088 Working Capital to Assets Ratio****089 Inverse PEG****090 1Y Chg in Sales to Price**

091 ● **Normalized ROA**092 ● **Capital Expenditure to Total Assets**093 ● **Consecutive Quarters of Declines in TTM Overhead/Sales**094 ● **Sales to Invested Capital**095 ● **26W Price Stat**096 ● **Inventory Turnover**

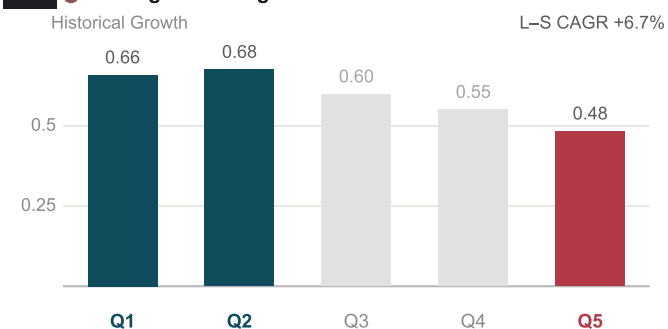
097 ● Year over Year Change in SGA to Sales



098 ● EBIT to Assets



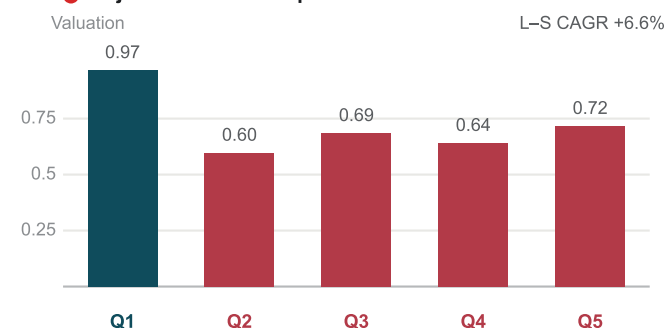
099 ● 1Y Chg in Earnings to Price



100 ● Long Term Debt to Equity Ratio

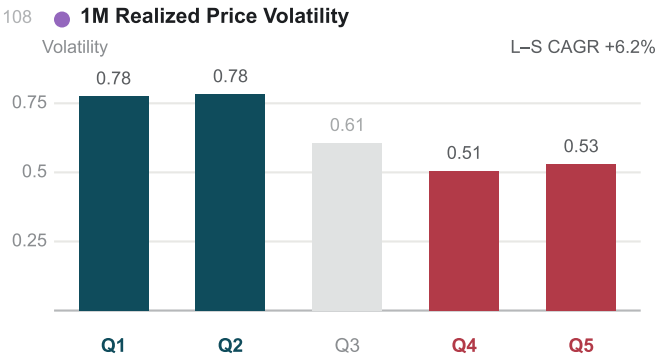
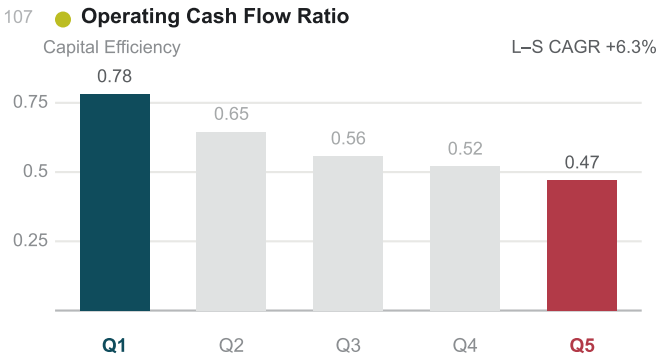
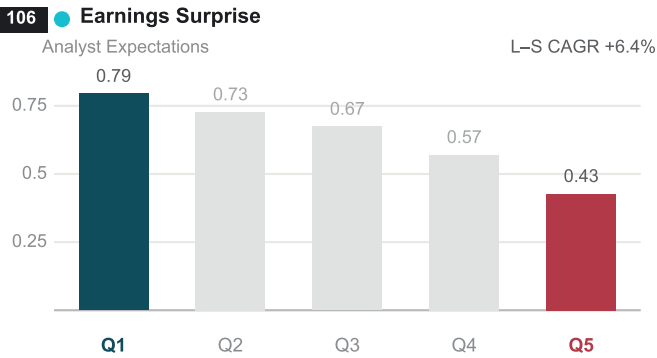
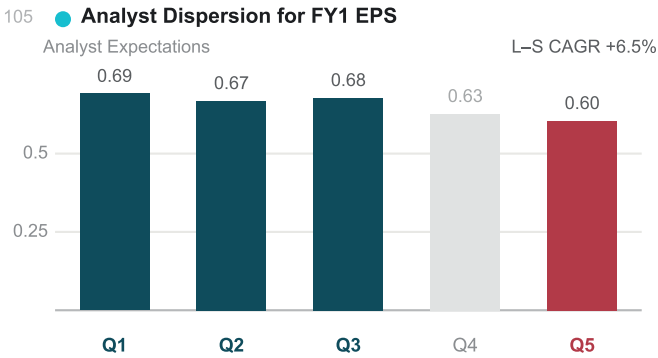
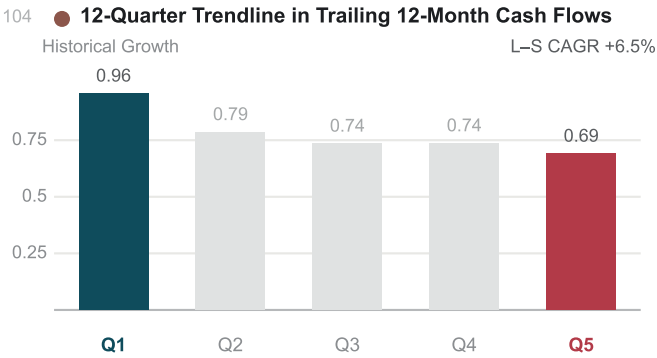
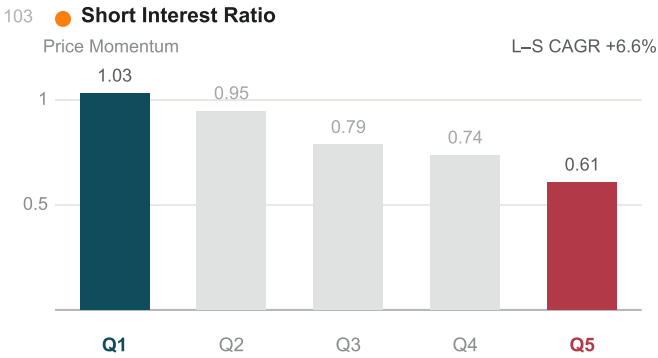


101 ● Adj. EBITDA to Enterprise Value



102 ● Adj 1Y Chg in Earnings per Share

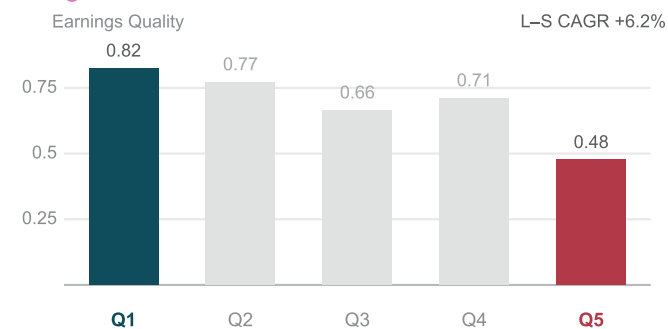




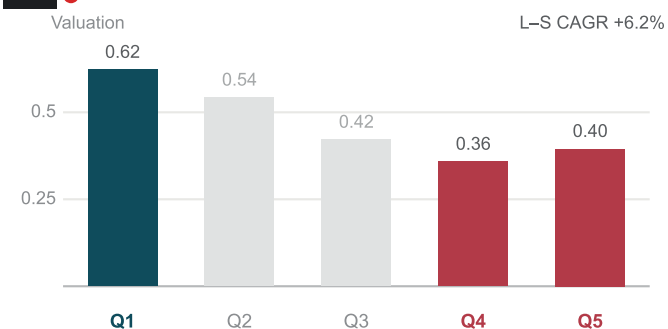
109 ● Volatility adjusted 12M return



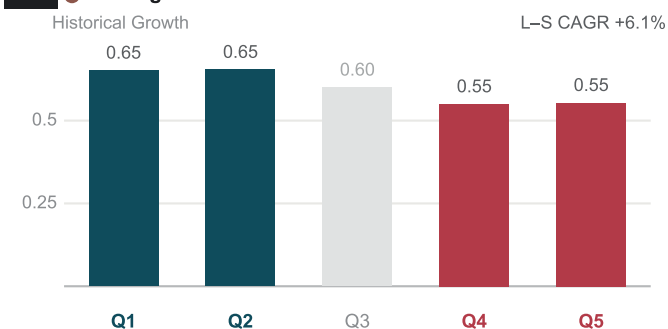
110 ● Current Ratio



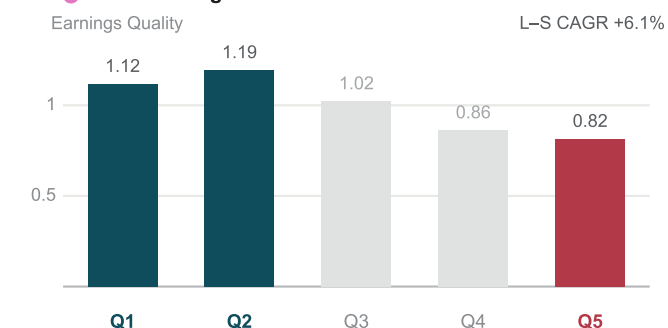
111 ● Cash to EV Ratio



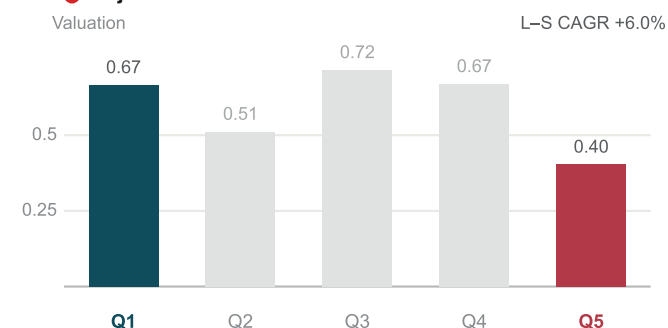
112 ● 1Y Chg in Cash Flow to Price

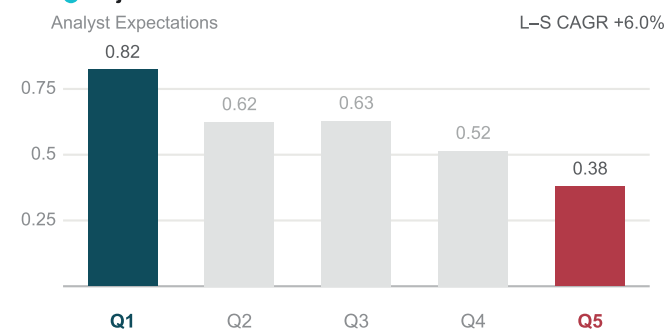
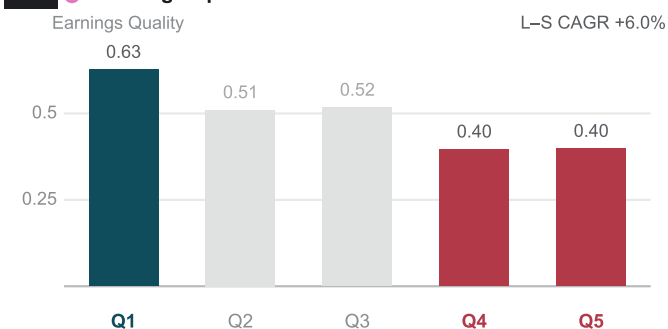
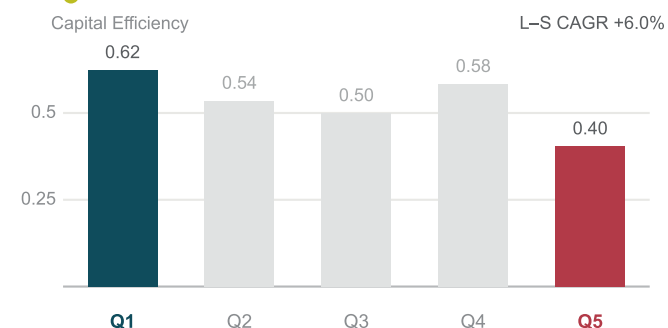


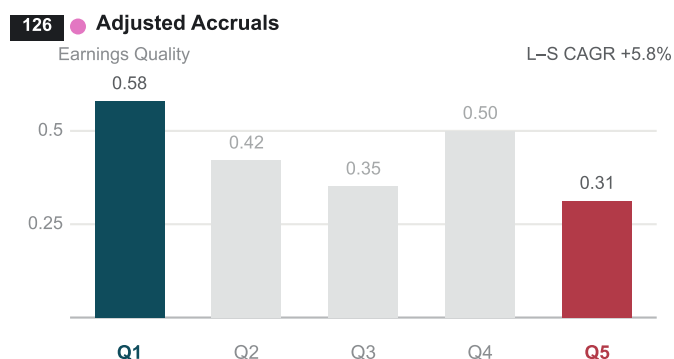
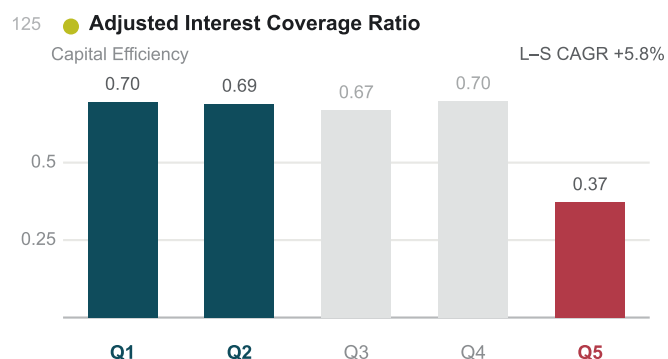
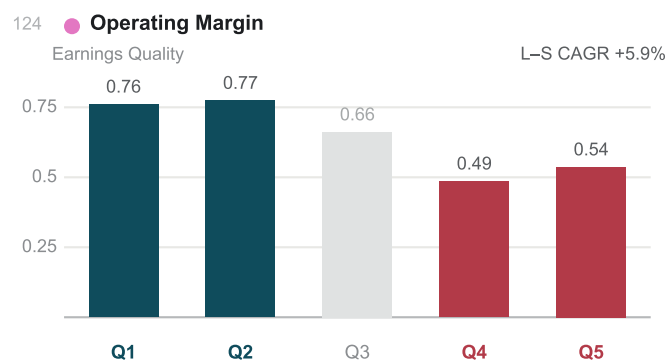
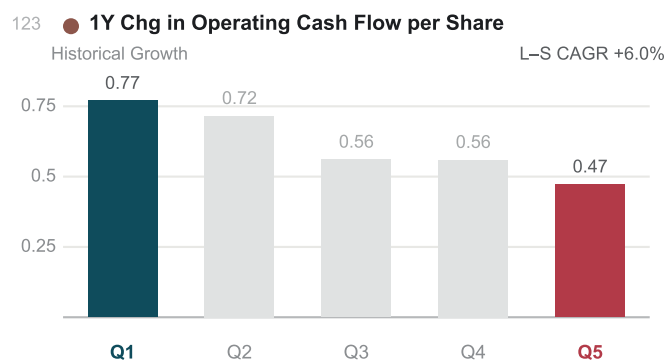
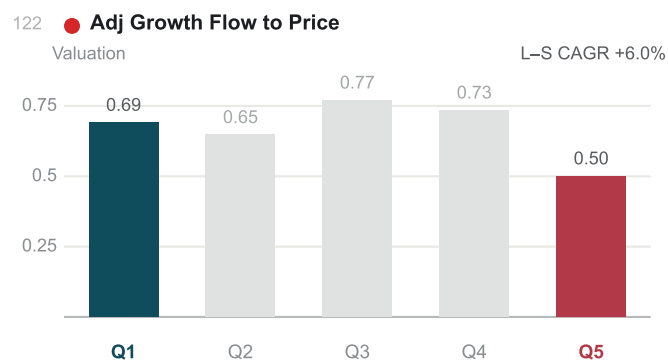
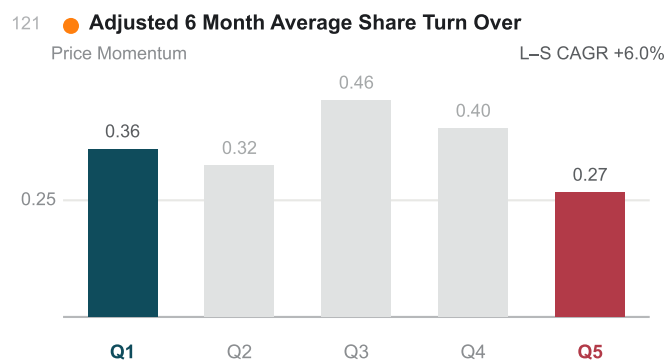
113 ● EBITDA Margin



114 ● Adj Free Cash Flow to Price



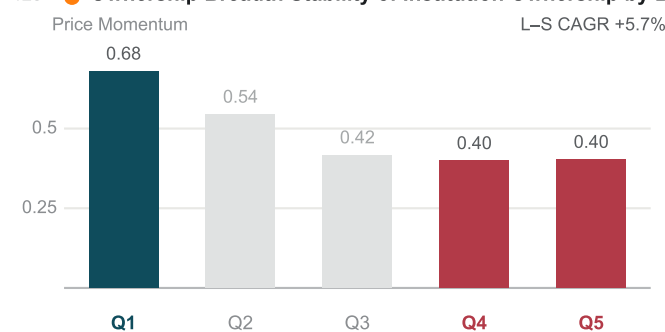
115 Capital Expenditure to Sales**116 Trailing 12-Month Overhead/Trailing 12-Month Sales****117 Adj Number of EPS FY2 Revisions****118 Working Capital Accruals****119 Asset Growth - 1 Year****120 3Y Chg in Operating Cash Flow per Share**



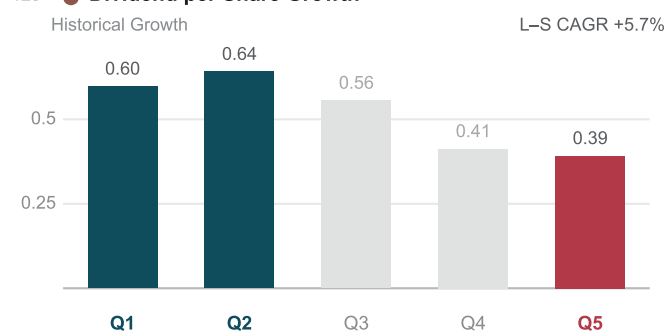
127 ● 3Y Chg in Free Cash Flow to Price



128 ● Ownership Breadth Stability of Institution Ownership by Buyers



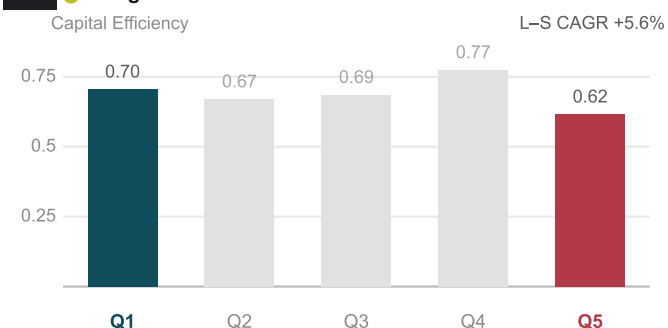
129 ● Dividend per Share Growth



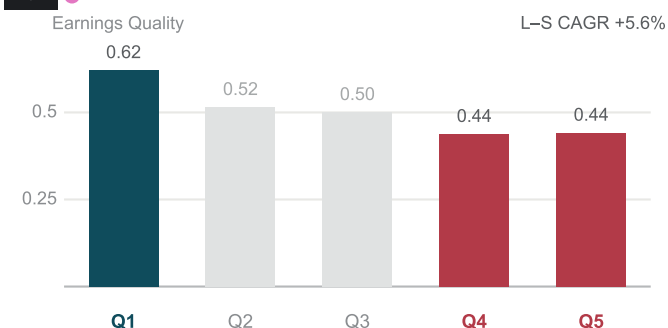
130 ● Accrual Ratio - Balance Sheet



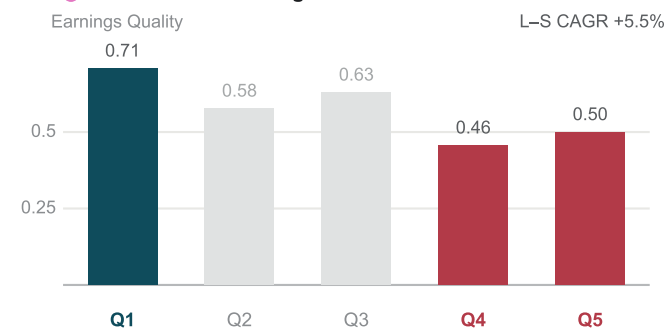
131 ● Long Term Debt to Assets Ratio



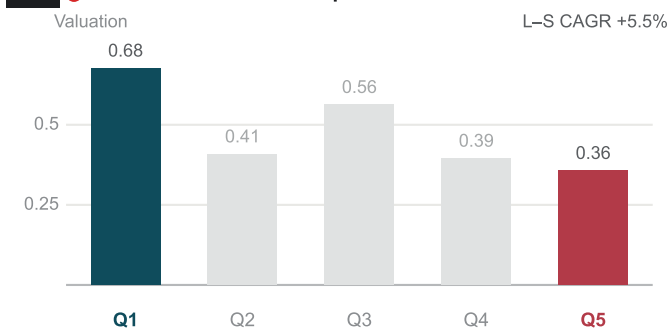
132 ● Accrual Ratio - Cash Flows



133 ● Year over Year Change of EPS to Sales



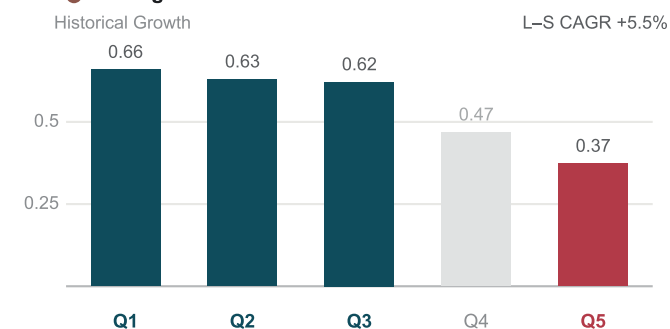
134 ● Free Cash Flow to Enterprise Value



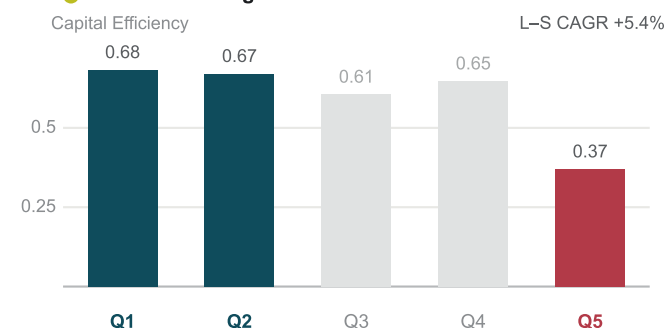
135 ● Operating Cash Flow to Asset



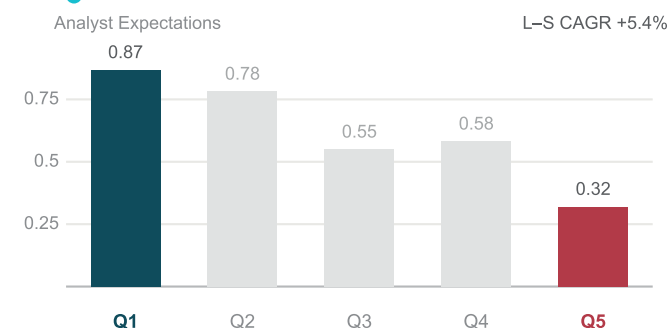
136 ● 3Y Chg in Sales to Price

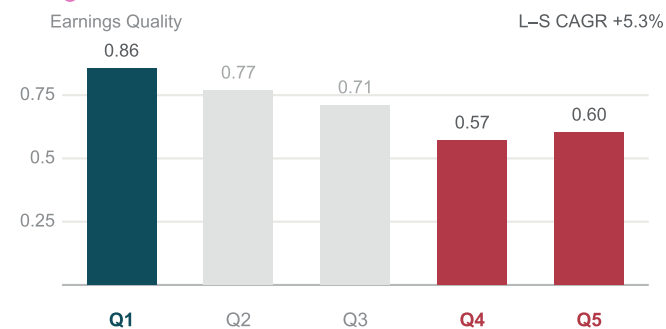
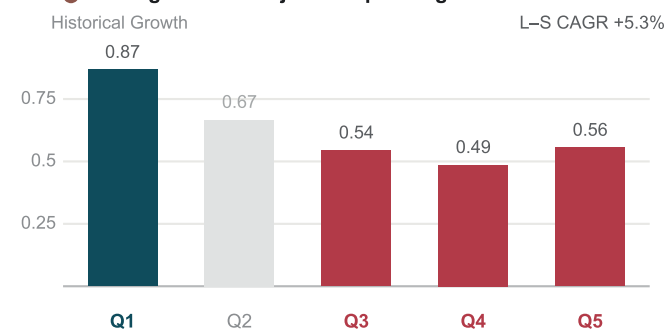
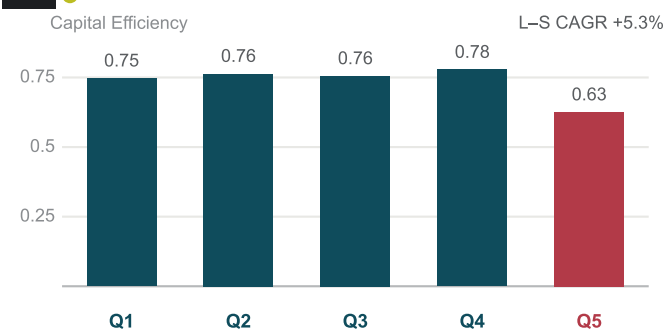
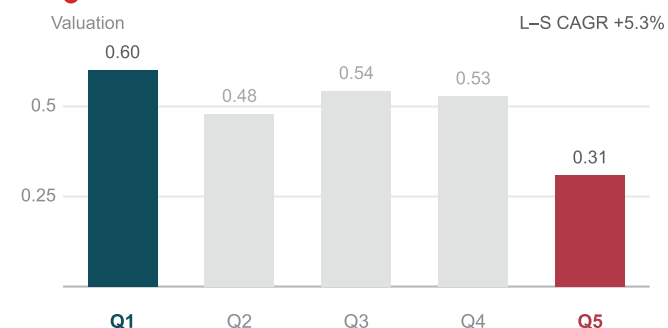


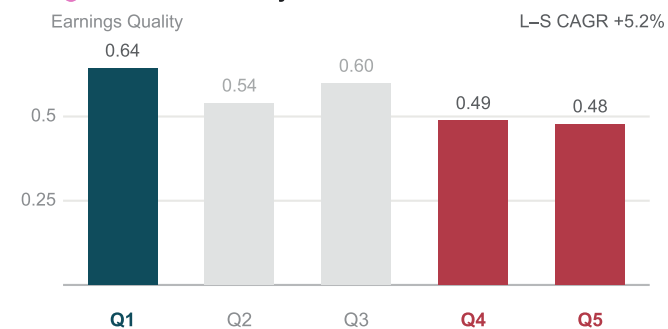
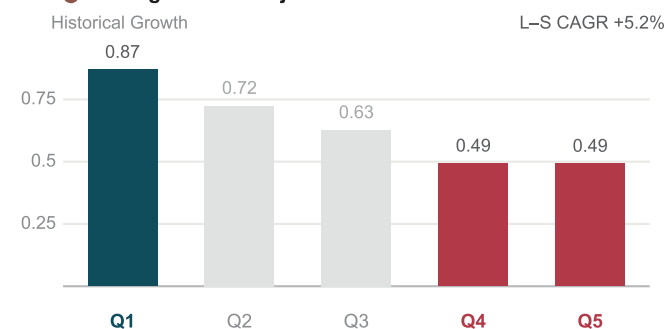
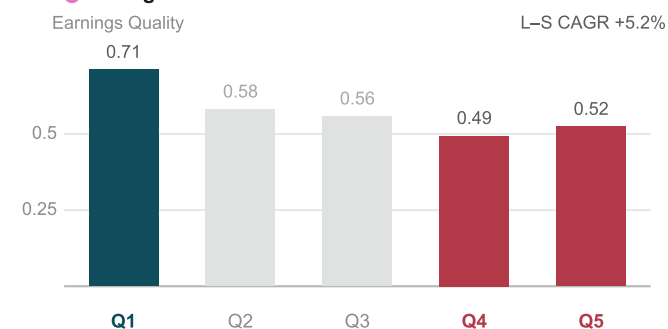
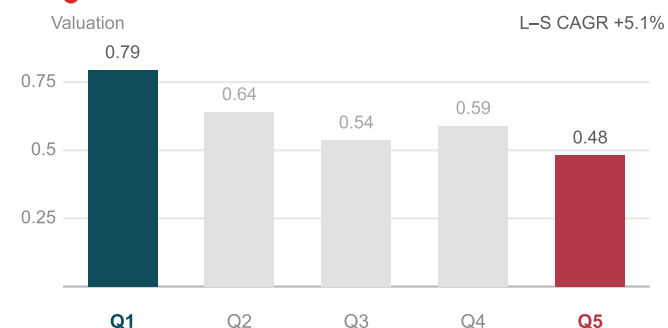
137 ● Interest Coverage Ratio

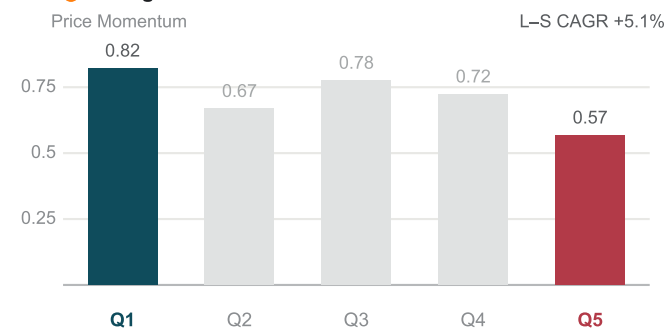
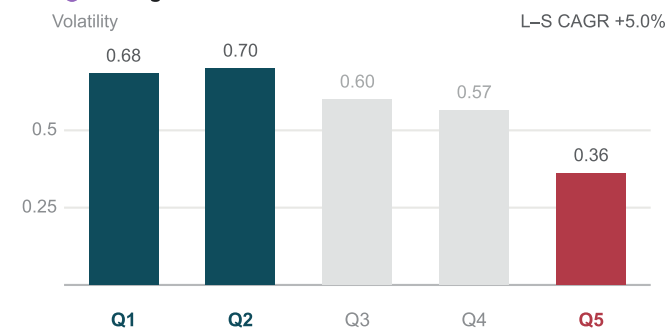
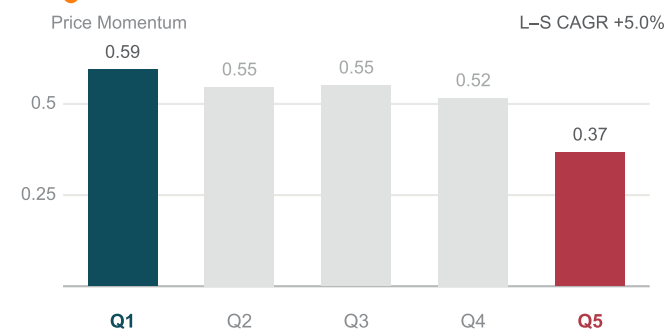
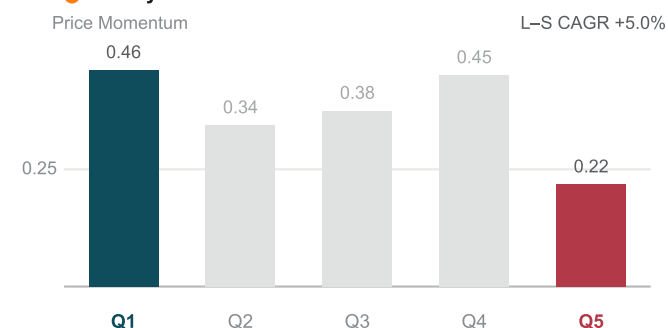


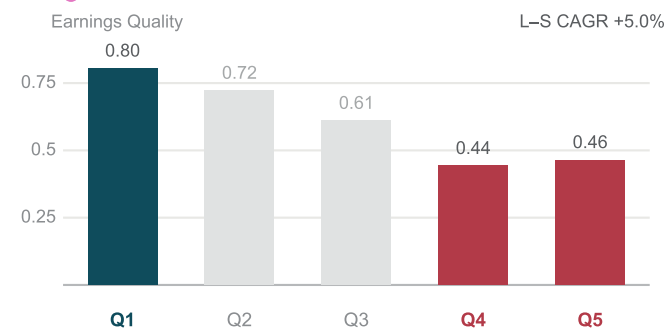
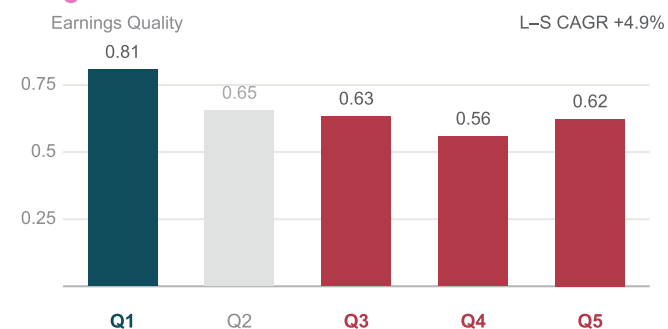
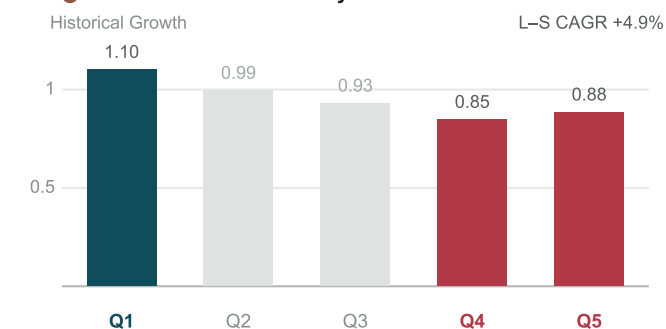
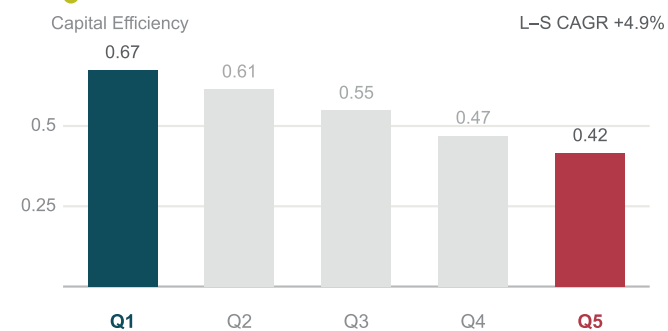
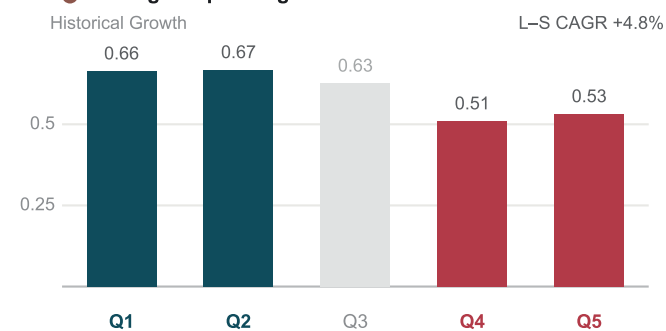
138 ● Number of EPS FY1 Revisions



139 ● **Num of Quarters of Declines in Receivables and Inventories/...**140 ● **Quick Ratio**141 ● **1Y Chg in Asset Adjusted Operating Cash Flow**142 ● **Debt to Assets Ratio**143 ● **Forward Free Cash Flow to Price Ratio**144 ● **Working Capital to Total Assets**

145 ● **Net Income Stability**146 ● **Return on Equity Stability**147 ● **1Y Chg in Asset Adjusted Free Cash Flow**148 ● **Change EPS to Sales**149 ● **Cash to Total Assets**150 ● **6M Momentum in Trailing 12M Sales**

151 ● **Change in Number of Holders**152 ● **1Y Chg in Amihud**153 ● **24M Residual Return Variance**154 ● **Closing Price to 52 Week High**155 ● **Free Cash Flow Stability**156 ● **30 Day Coefficient of Variation of Volume to Price**

157 ● **Cash Ratio**158 ● **3Y Chg in Asset Adjusted Operating Cash Flow**159 ● **SG&A to Sales**160 ● **Return on Assets Stability**161 ● **Receivables Turnover Ratio**162 ● **1Y Chg in Operating Cash Flow to Price**

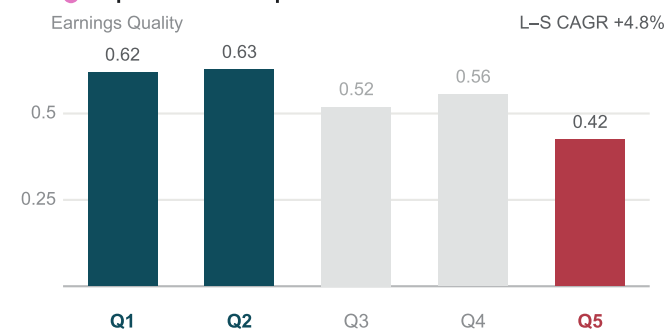
163 ● 1Y Chg in Shares Outstanding



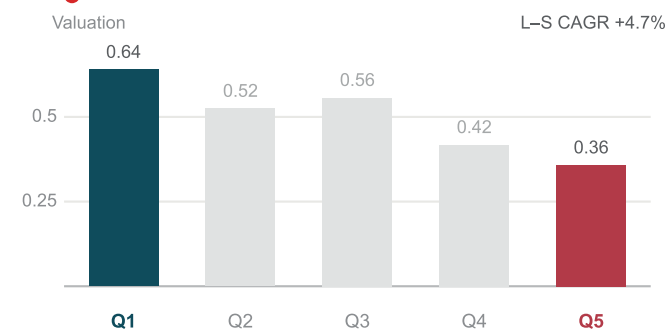
164 ● 1Y Chg in Stock Buybacks



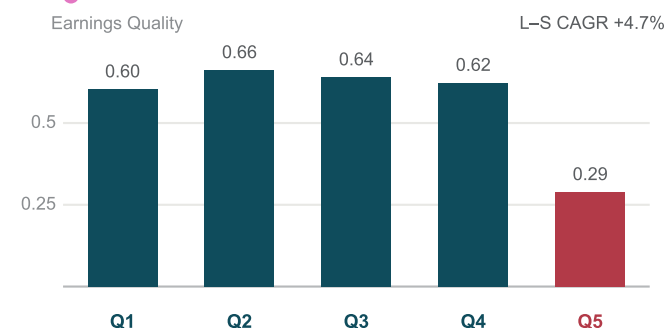
165 ● Depreciation to Capex Ratio



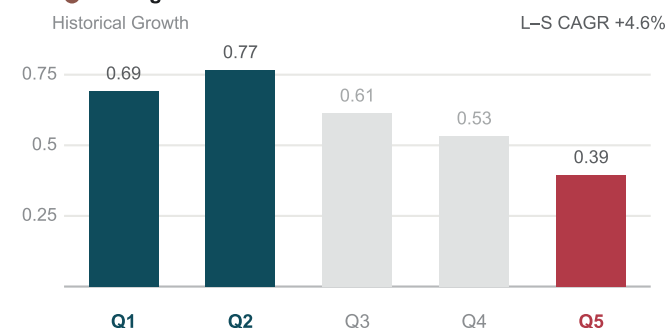
166 ● Free Cash Flow to Price

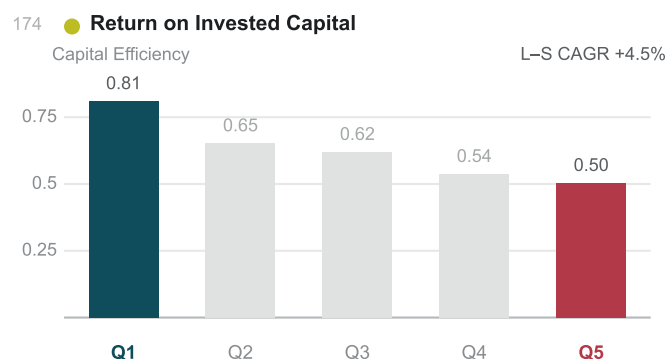
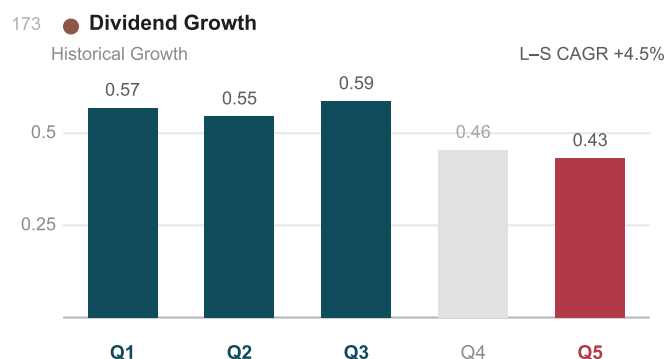
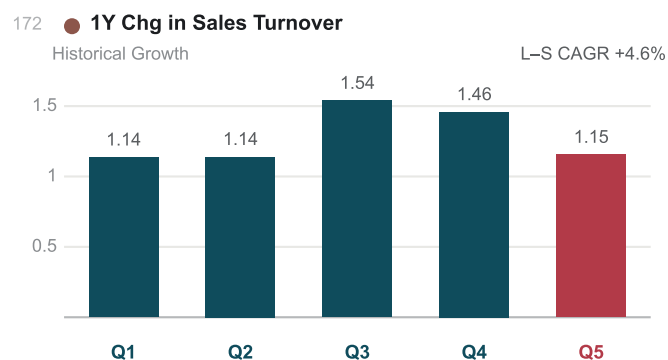
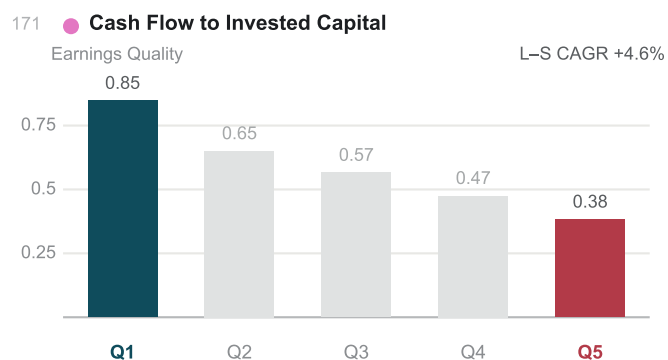
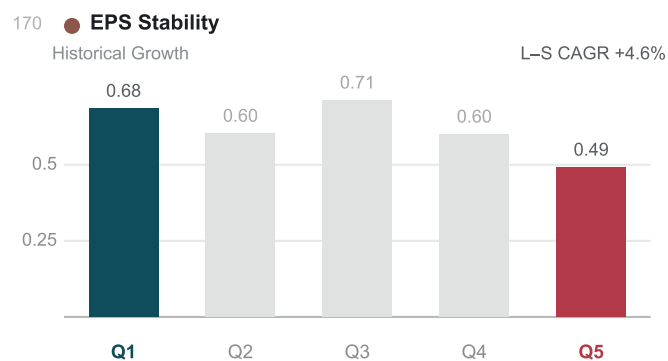
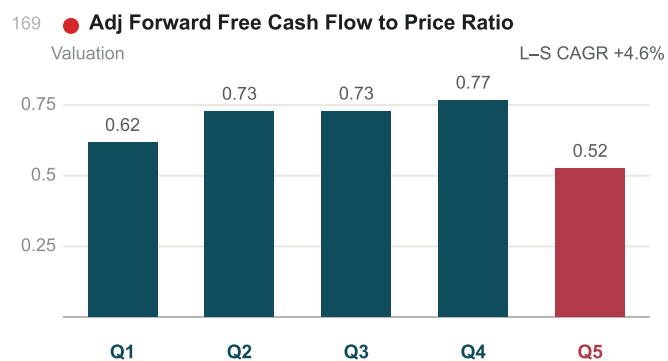


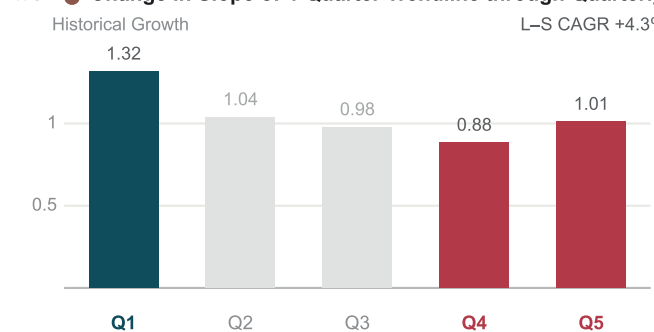
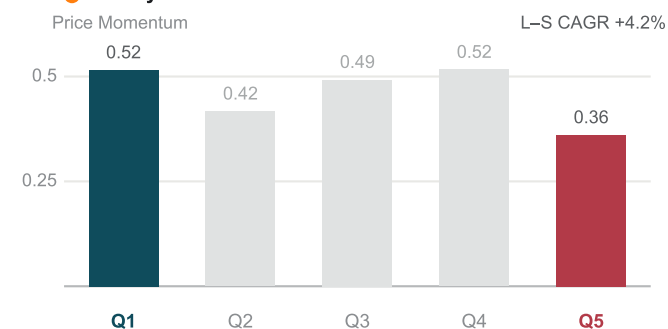
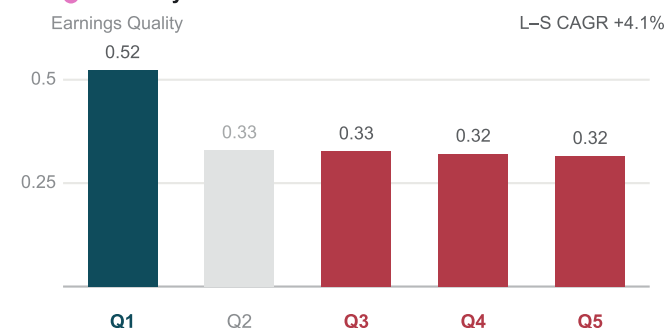
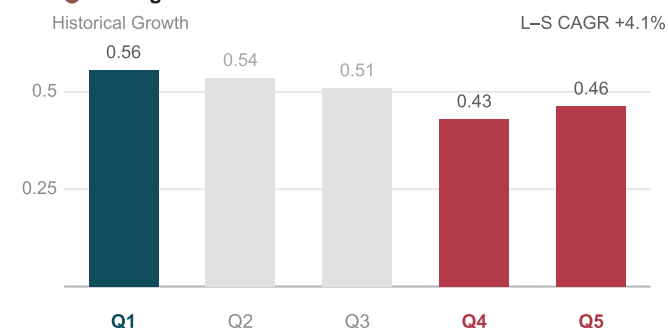
167 ● Asset Turnover

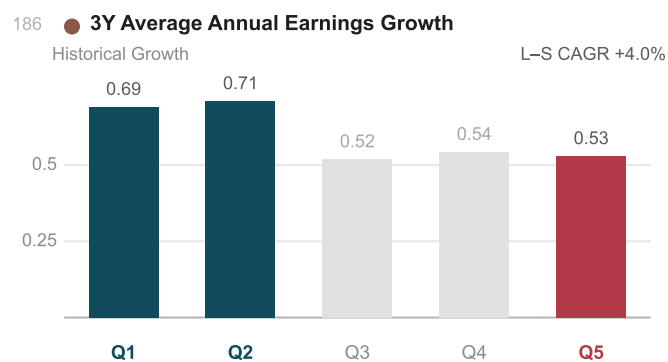
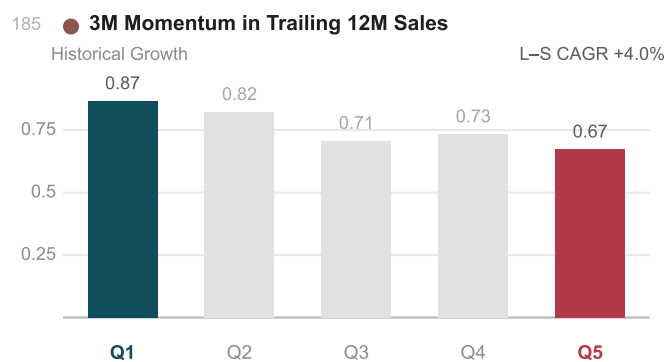
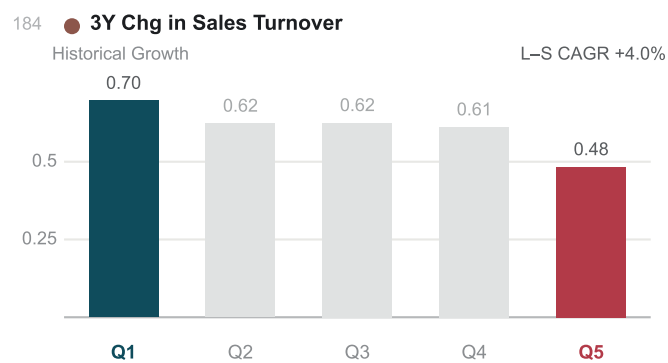
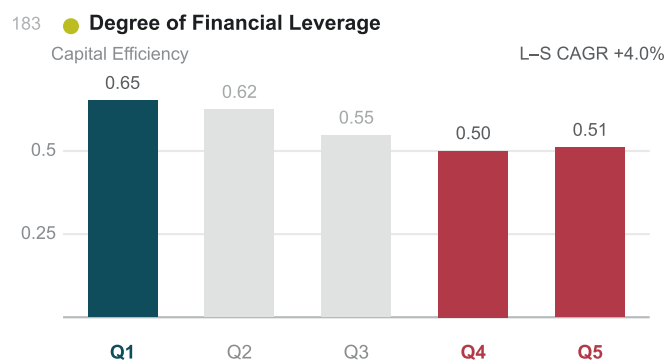
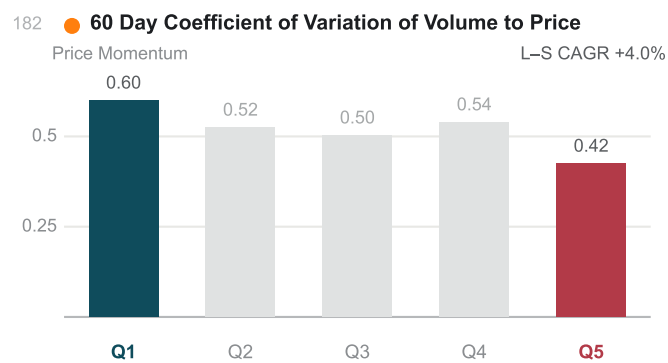
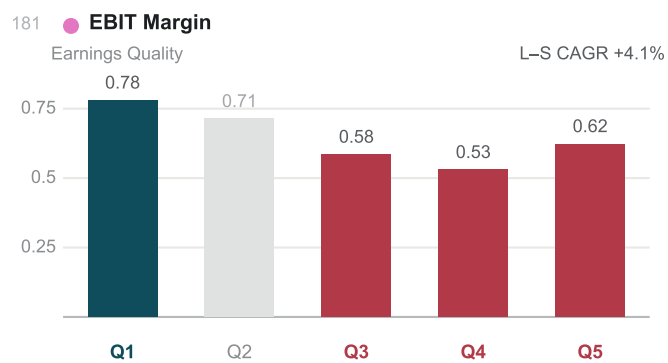


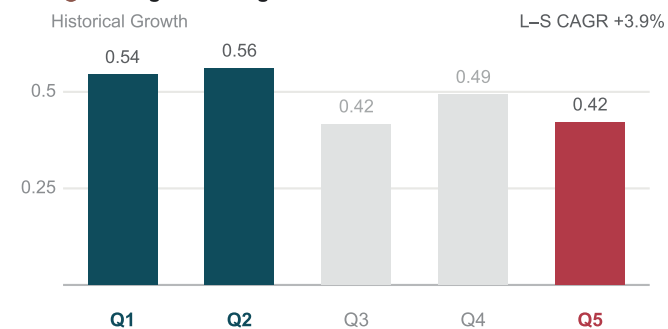
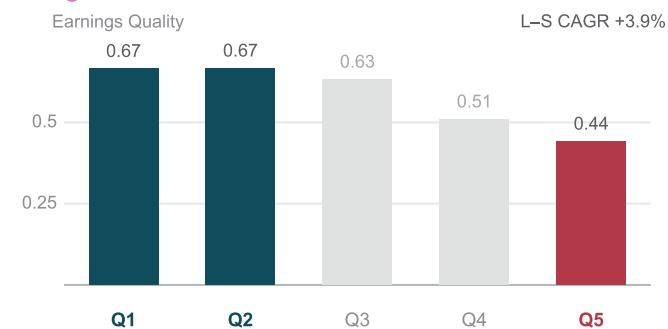
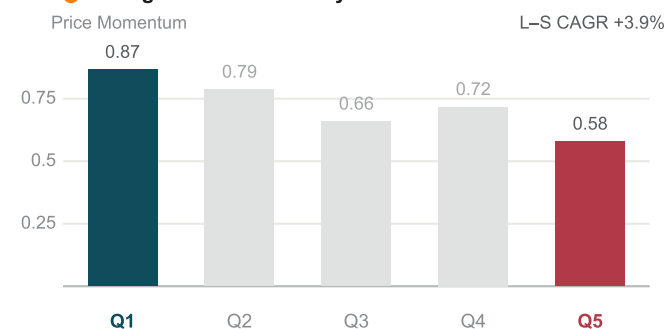
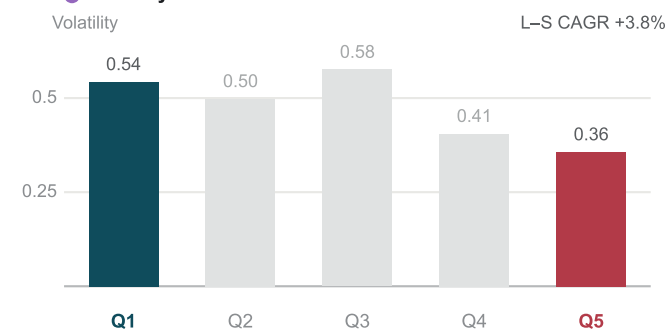
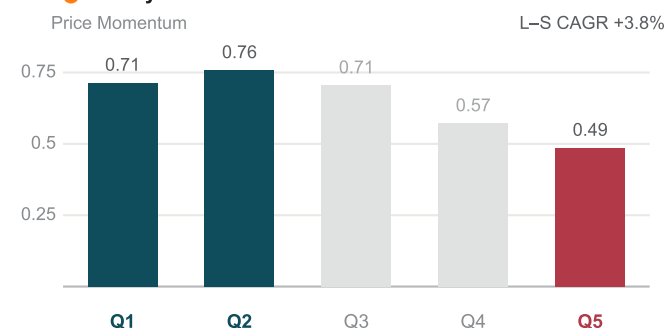
168 ● 1Y Chg in Free Cash Flow to Price

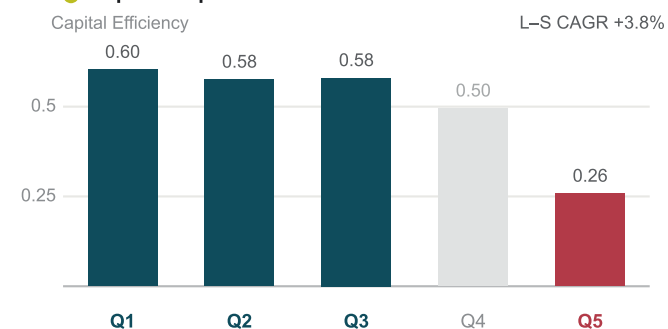
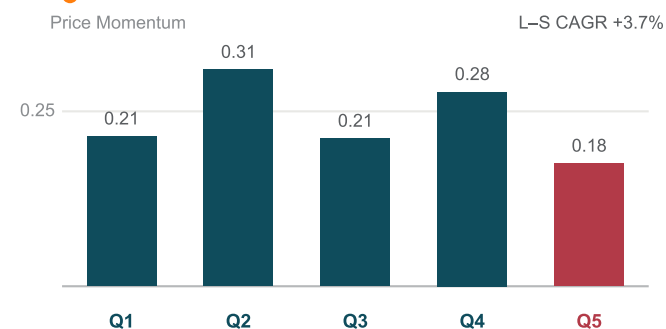
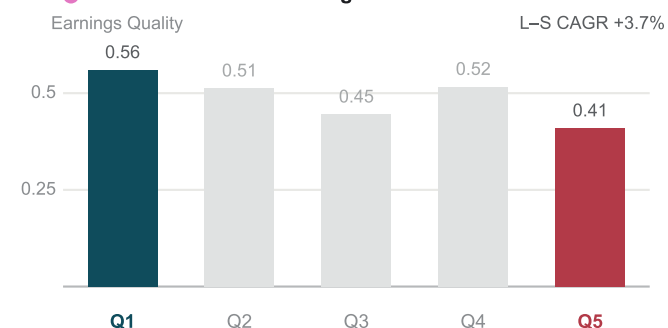


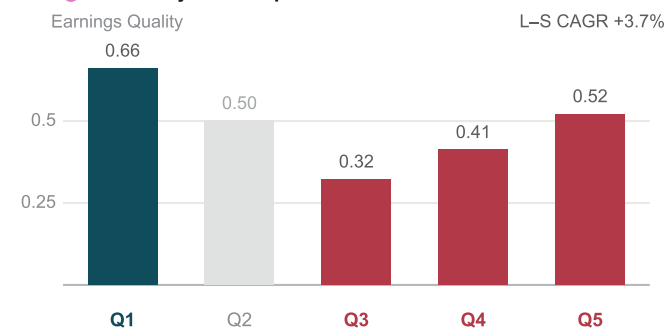
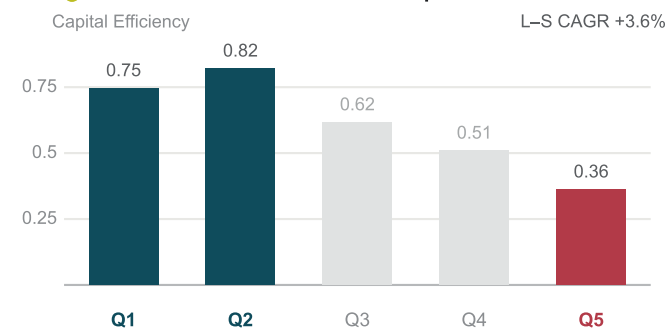
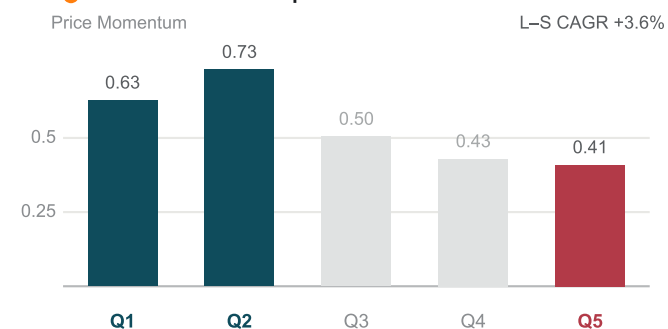
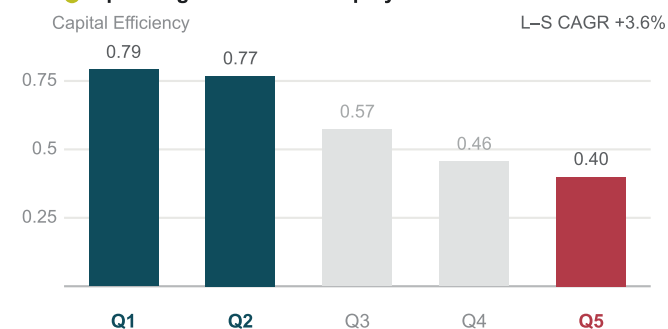
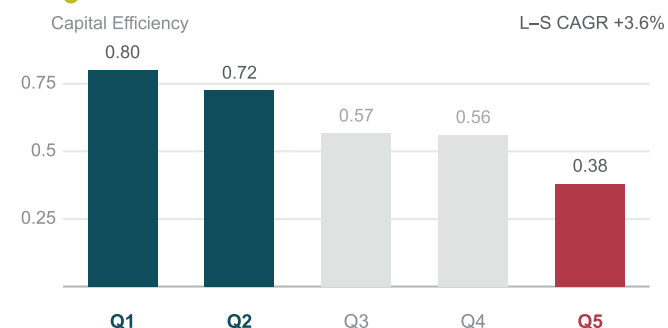
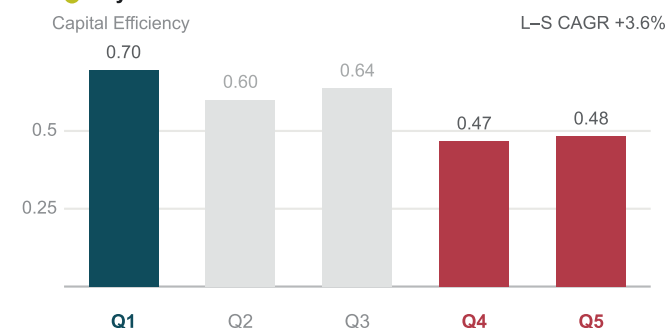


175 ● **Standardized Unexpected Earnings**176 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**177 ● **Cash Burn Rate**178 ● **20 Day Coefficient of Variation of Volume to Price**179 ● **Inventory to Assets Ratio**180 ● **3Y Chg in Cash Flow to Price**

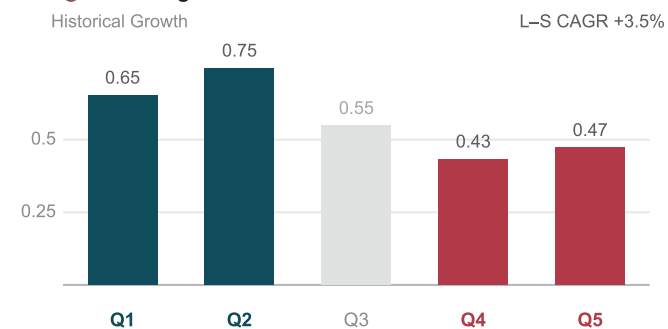


187 ● **3Y Chg in Earnings to Price**188 ● **Free Cash Flow to Sales**189 ● **Change in Number of Buyers**190 ● **130 Day Minimum Return**191 ● **10 Day MACD Trend**192 ● **ROA 60 Month Slope**

193 ● **Cash Conversion Cycle**194 ● **Net Profit Margin**195 ● **Capital Acquisition Ratio**196 ● **Share Turnover**197 ● **Sales to Gross Profit Margin Growth**198 ● **Research & Development Intensity**

199 ● **Asset Adjusted Capital Investments**200 ● **Cash Flow Return on Invested Capital**201 ● **Institution Ownership Level**202 ● **Operating Cash Flow to Equity**203 ● **Gross Profit to Assets**204 ● **Payout Ratio**

205 ● 3Y Average Annual Sales Growth



206 ● 3Y Chg in Asset Adjusted EPS



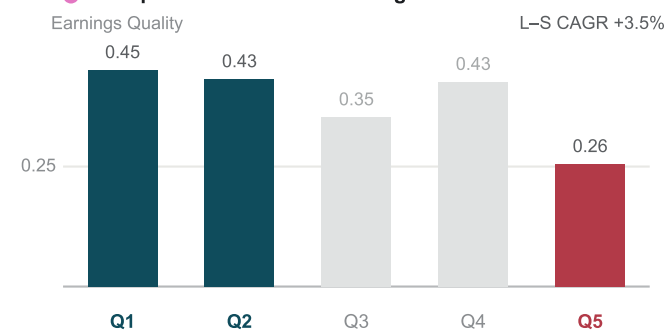
207 ● 3Y Chg in Free Cash Flow per Share



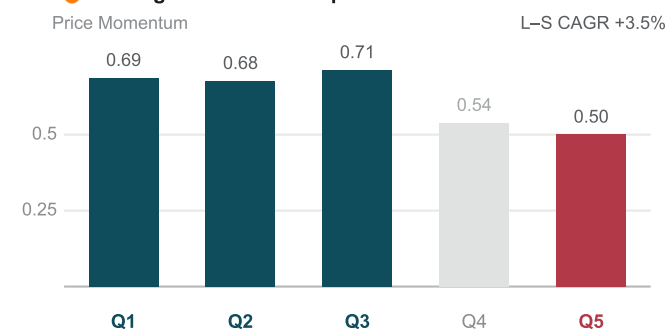
208 ● 1Y Chg in Earnings per Share



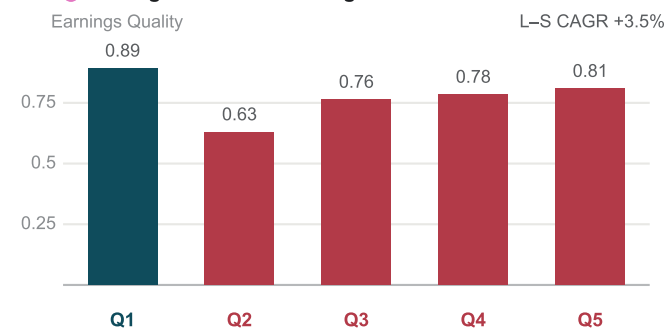
209 ● Unexpected Receivables Change



210 ● 6M Chg in 36M CAPM Alpha



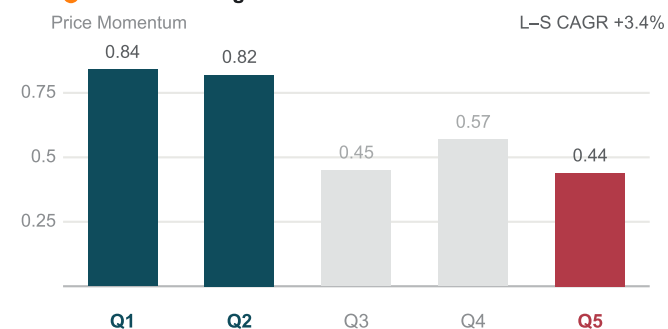
211 ● 1Y Chg in Sales to Earnings



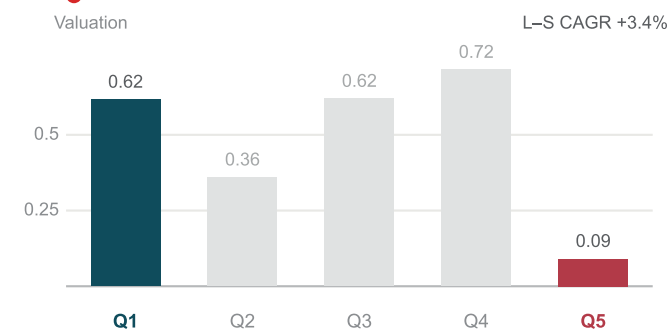
212 ● Operating Cash Flow Stability



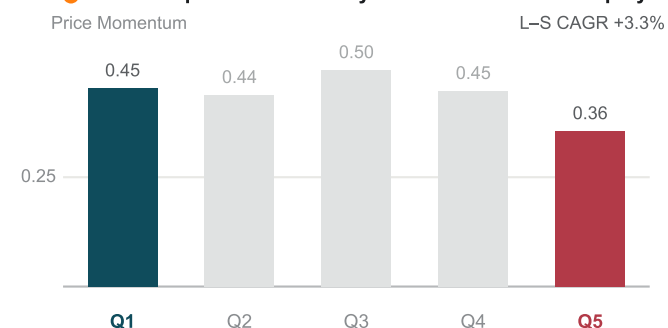
213 ● 6 Month Average Share Turn Over



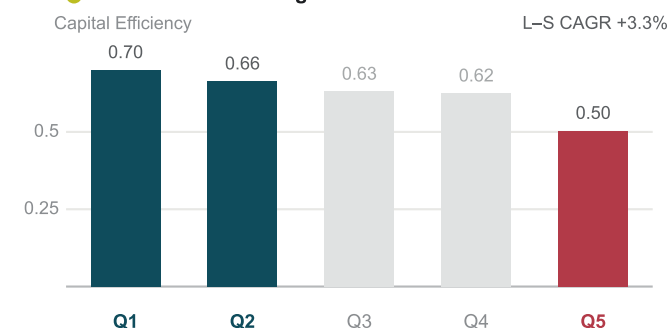
214 ● EBIT to Price



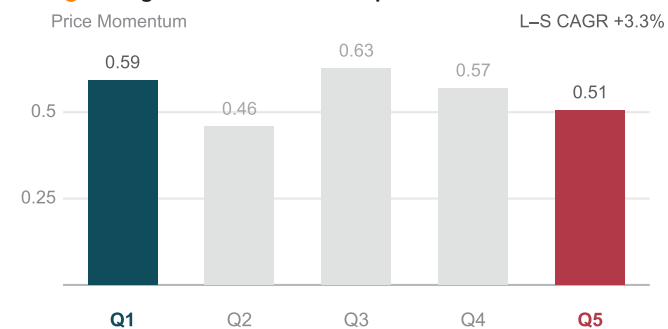
215 ● Ownership Breadth Stability of Institution Ownership by Hol...



216 ● Year over Year Change of Total Debt



217 ● Foreign Institution Ownership



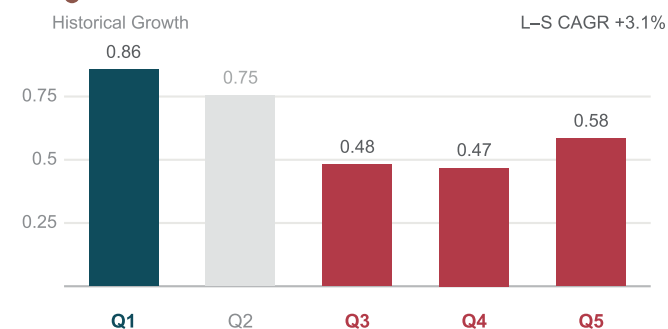
218 ● 3Y Chg in Asset Adjusted Free Cash Flow



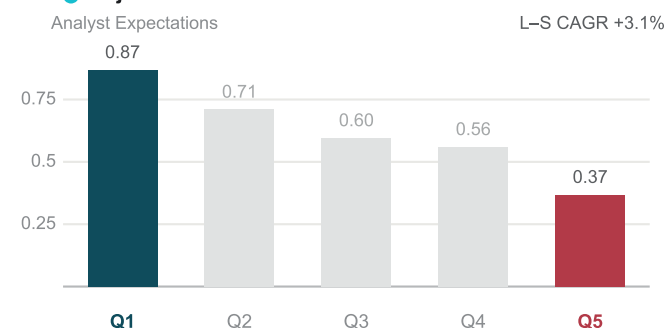
219 ● Free Cash Flow to Common Equity



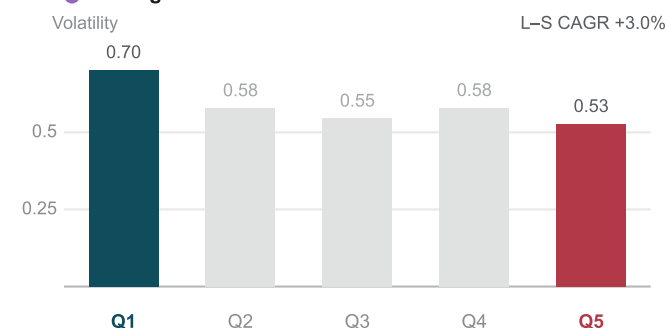
220 ● Sustainable Growth Rate



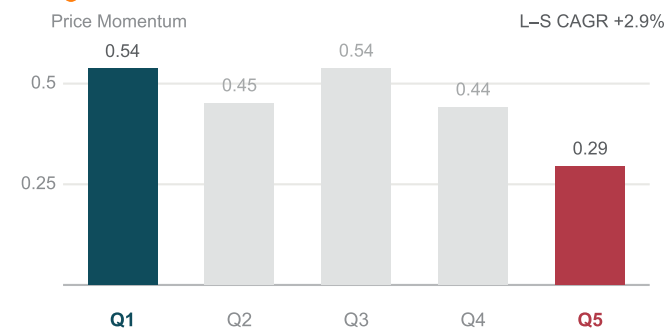
221 ● Adjusted Number of EPS FY1 Revisions



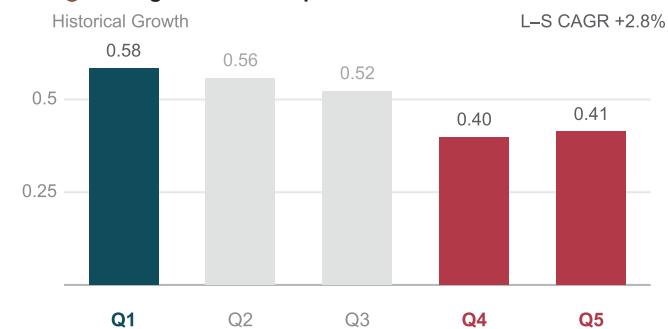
222 ● 1Y Chg in Debt to Assets



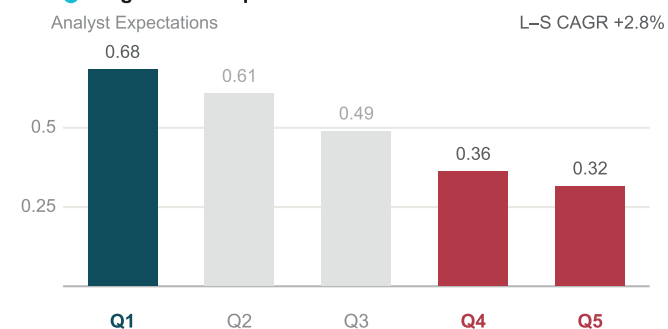
223 ● Standardized 36M Residual



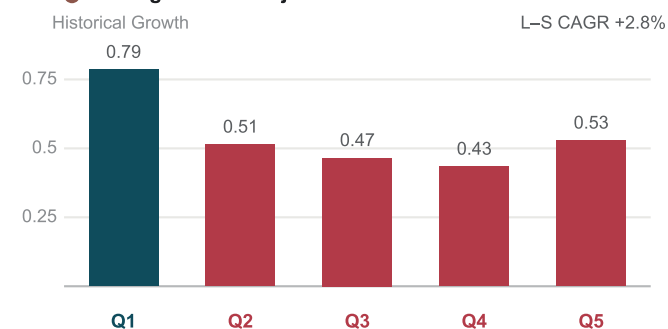
224 ● 1Y Chg in Cash Flow per Share



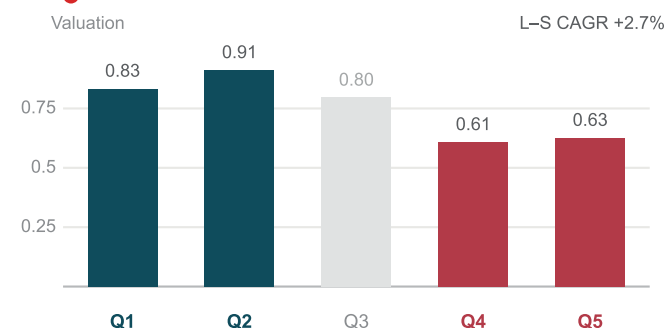
225 ● Target Price Gap to 6M EMA



226 ● 3Y Chg in Asset Adjusted Cash Flow

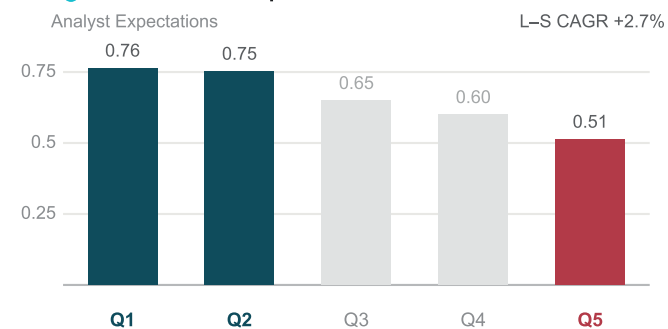
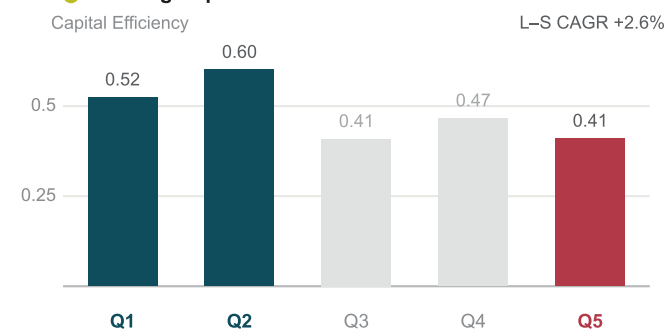


227 ● Net Current Assets to Price Ratio

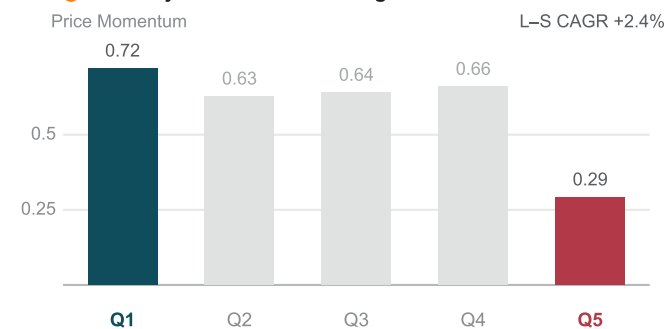


228 ● Pretax Income to Net Operating Assets



229 ● **EBIT Estimate Dispersion**230 ● **1Y Chg in EPS to Operating Cash Flow**231 ● **Working Capital Turnover Ratio**232 ● **Change of Sales to Change of EPS**233 ● **Pretax Profit Margin**234 ● **Sales Acceleration**

235 ● Risk Adjusted Relative Strength



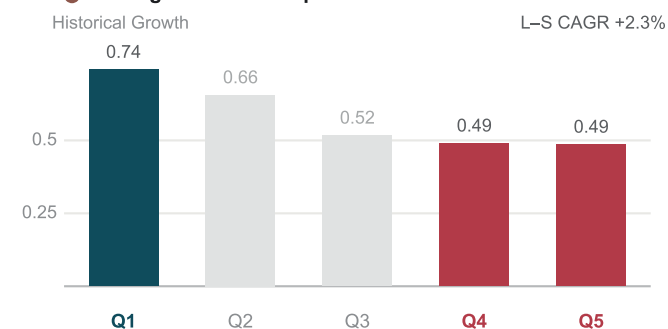
236 ● Analyst Earnings Estimate Diffusion



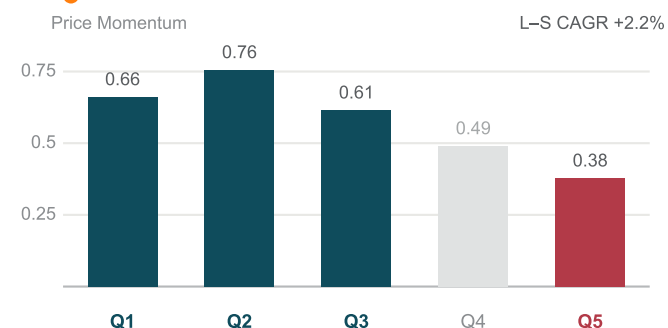
237 ● Change of Ownership Level - Hedge Fund



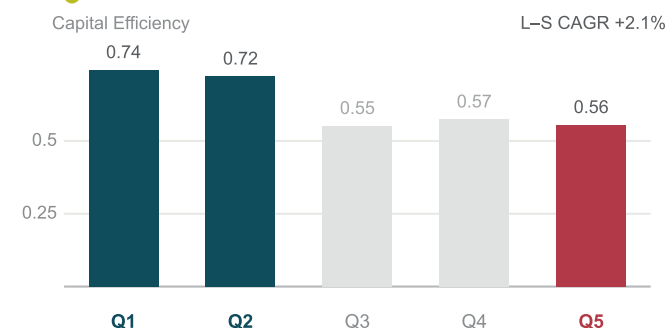
238 ● 3Y Chg in Cash Flow per Share

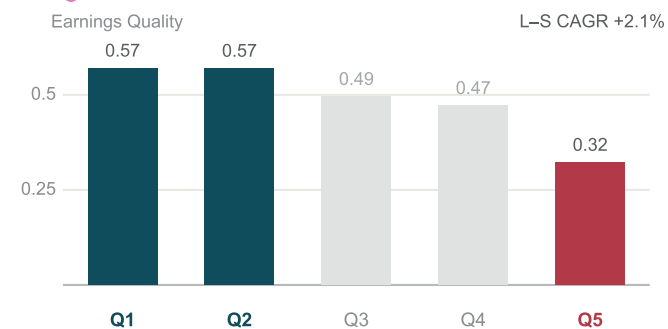
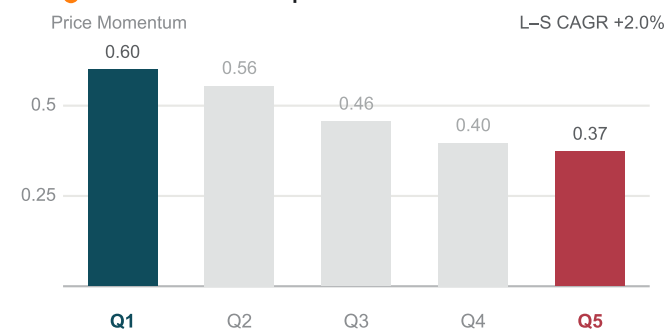
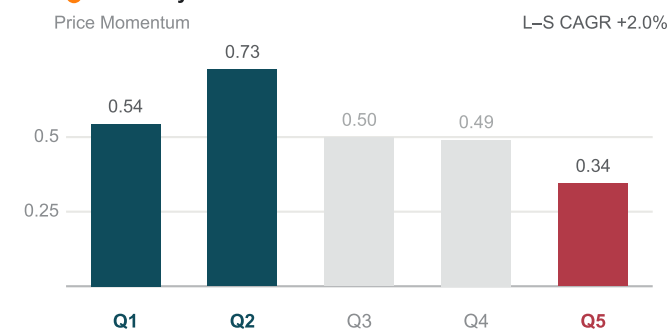
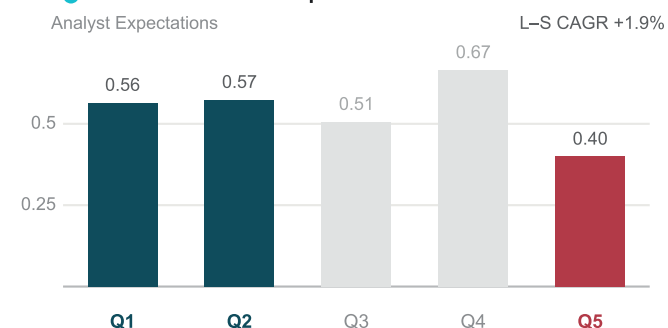
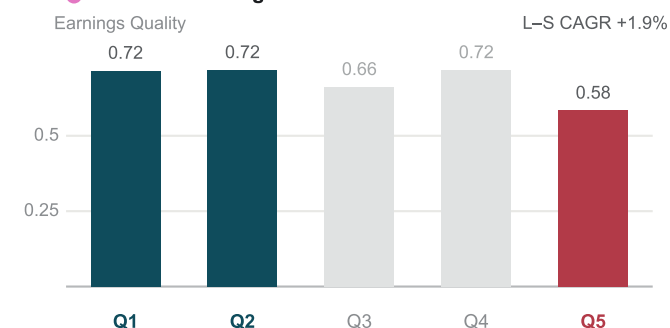


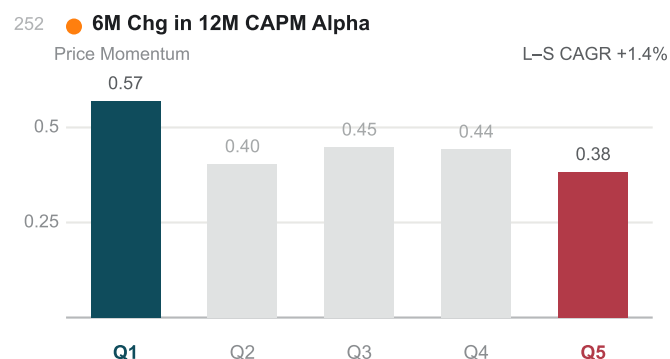
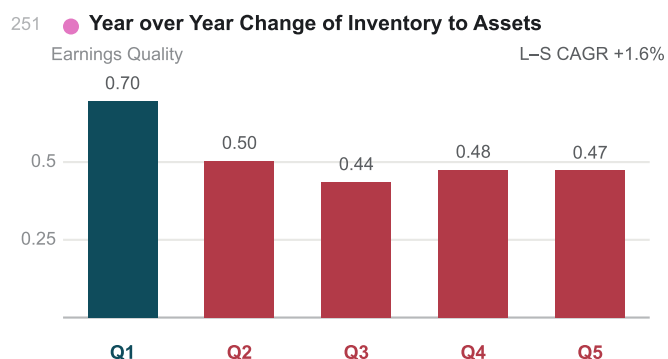
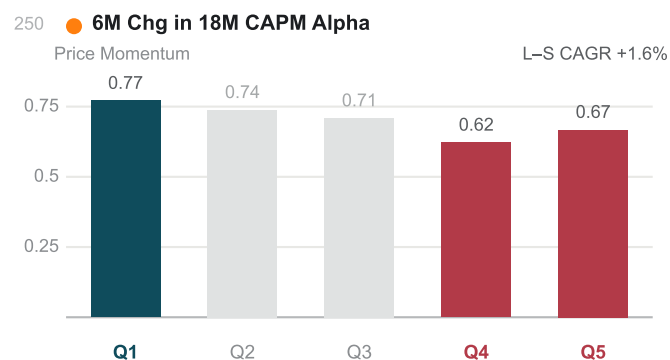
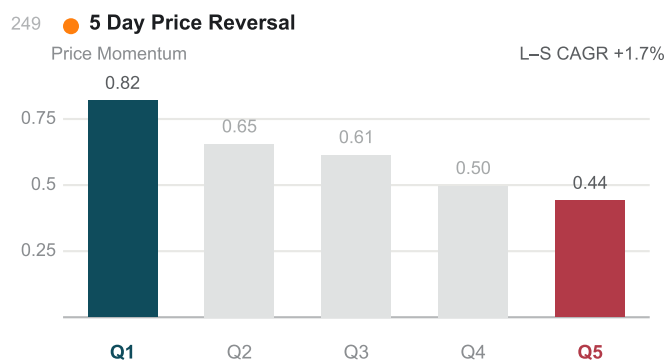
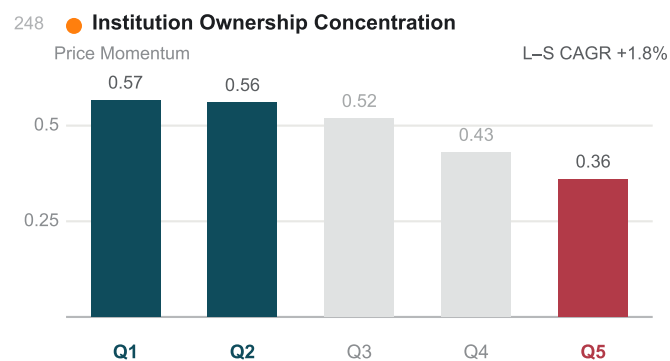
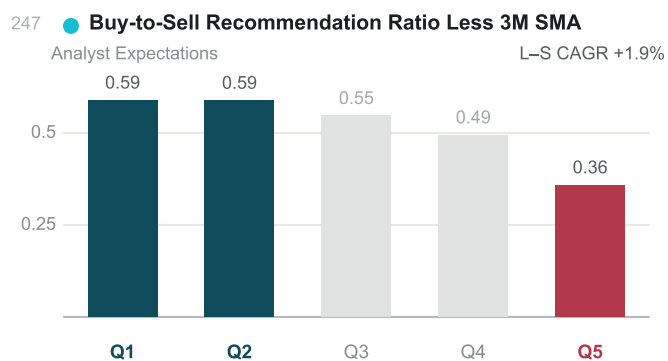
239 ● Investment Duration



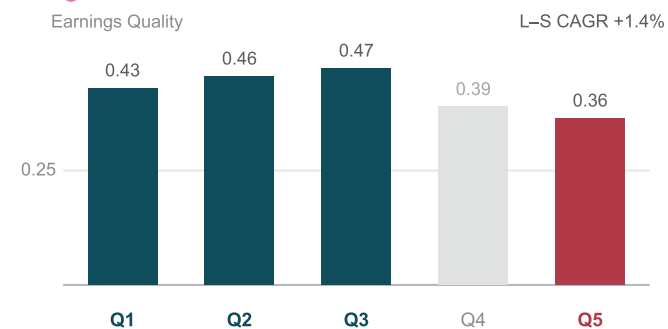
240 ● Normalized ROE



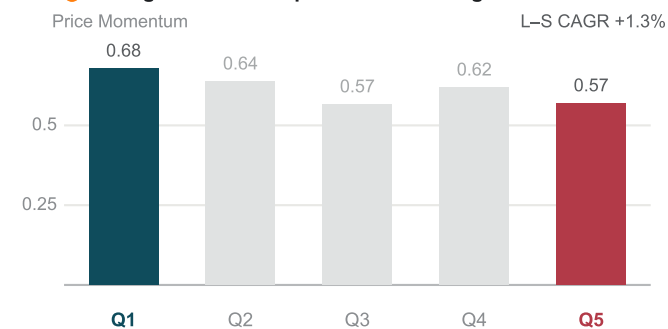
241 ● **Cash to Sales**242 ● **3Y Chg in Earnings per Share**243 ● **Institution Ownership Turnover**244 ● **Max Daily Return in the Past 6 Months**245 ● **Revenue Estimate Dispersion**246 ● **Retained Earnings to Total Assets**



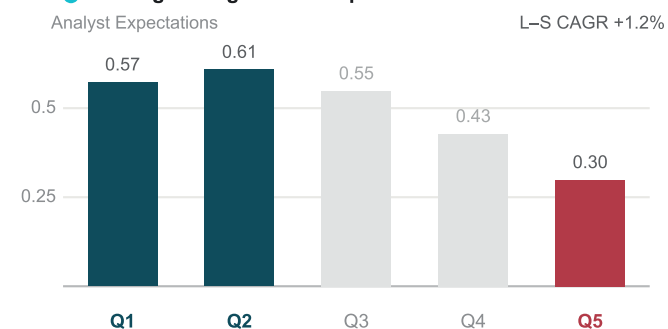
253 ● ROE 20 Qtr Standard Deviation



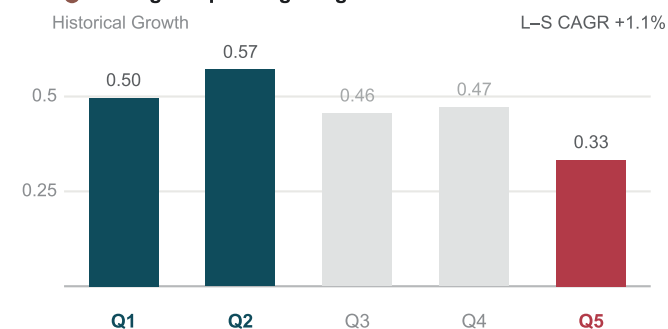
254 ● Change of Ownership Level - All Managers



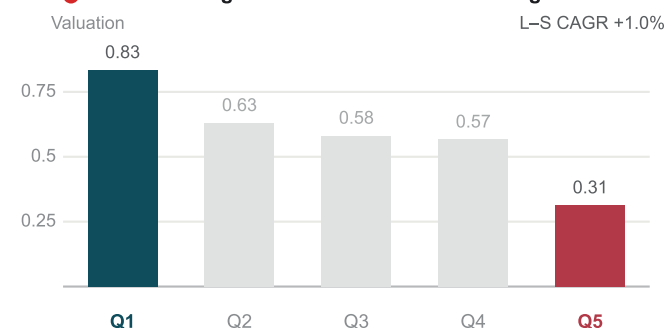
255 ● 6M Chg in Target Price Gap



256 ● 3Y Chg in Operating Margin

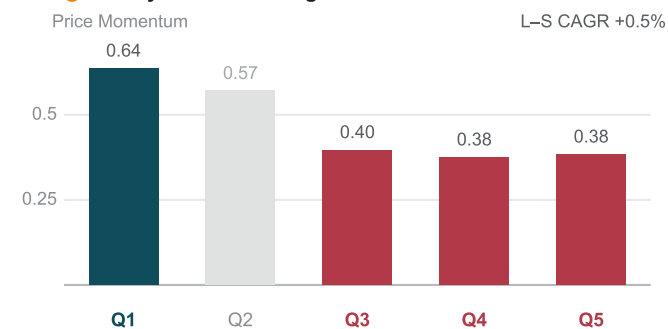
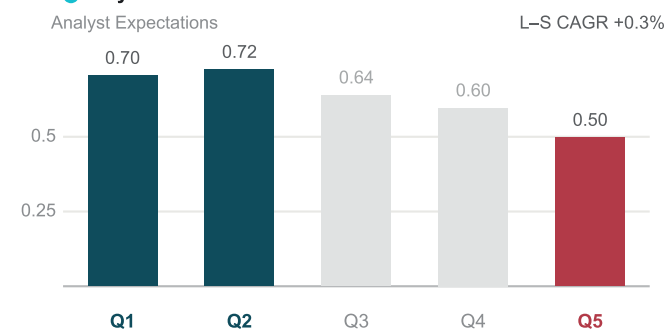
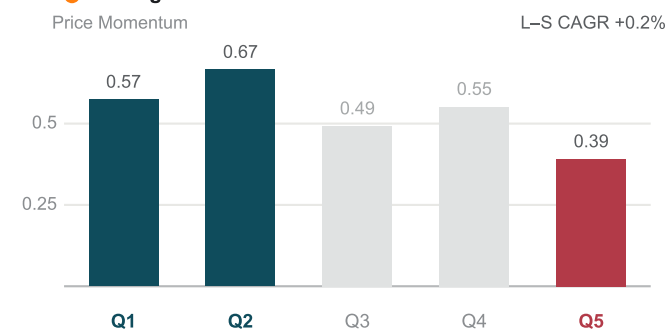
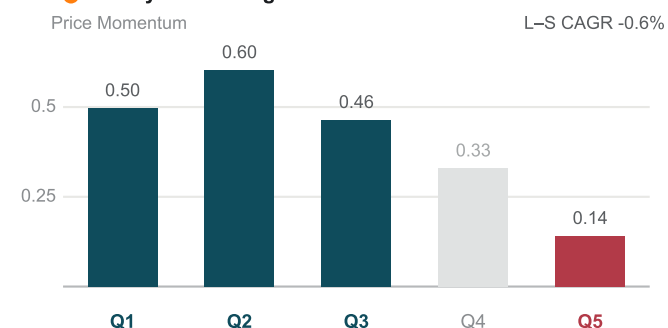


257 ● 8-Week Change in 12-Month Forward Earnings Consensus Estimate



258 ● Change in TTM Depr. to Capex



259 ● **6M Avg Chg 1M Recommendation**260 ● **14-Day Relative Strength Index**261 ● **Buy-to-Sell Recommendation Ratio Less 3M EMA**262 ● **1Y Chg in Share Turnover**263 ● **50 Day Volume Signal**264 ● **Adj 50 Day Volume Signal**